1 INTRODUCTION

In which we try to explain why we consider artificial intelligence to be a subject most worthy of study, and in which we try to decide what exactly it is, this being a good thing to decide before embarking.

We call ourselves *Homo sapiens*—man the wise—because our mental capacities are so important to us. For thousands of years, we have tried to understand *how we think*; that is, how a mere handful of stuff can perceive, understand, predict, and manipulate a world far larger and more complicated than itself. The field of **artificial intelligence**, or AI, goes further still: it attempts not just to understand but also to *build* intelligent entities.

ARTIFICIAL INTELLIGENCE

AI is one of the newest sciences. Work started in earnest soon after World War II, and the name itself was coined in 1956. Along with molecular biology, AI is regularly cited as the "field I would most like to be in" by scientists in other disciplines. A student in physics might reasonably feel that all the good ideas have already been taken by Galileo, Newton, Einstein, and the rest. AI, on the other hand, still has openings for several full-time Einsteins.

AI currently encompasses a huge variety of subfields, ranging from general-purpose areas, such as learning and perception to such specific tasks as playing chess, proving mathematical theorems, writing poetry, and diagnosing diseases. AI systematizes and automates intellectual tasks and is therefore potentially relevant to any sphere of human intellectual activity. In this sense, it is truly a universal field.

1.1 WHAT IS AI?

We have claimed that AI is exciting, but we have not said what it *is*. Definitions of artificial intelligence according to eight textbooks are shown in Figure 1.1. These definitions vary along two main dimensions. Roughly, the ones on top are concerned with *thought processes* and *reasoning*, whereas the ones on the bottom address *behavior*. The definitions on the left measure success in terms of fidelity to *human* performance, whereas the ones on the right measure against an *ideal* concept of intelligence, which we will call **rationality**. A system is rational if it does the "right thing," given what it knows.

RATIONALITY

Systems that think like humans	Systems that think rationally
"The exciting new effort to make computers think . . . *machines with minds*, in the full and literal sense." (Haugeland, 1985)	"The study of mental faculties through the use of computational models." (Charniak and McDermott, 1985)
"[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . ." (Bellman, 1978)	"The study of the computations that make it possible to perceive, reason, and act." (Winston, 1992)
Systems that act like humans	**Systems that act rationally**
"The art of creating machines that perform functions that require intelligence when performed by people." (Kurzweil, 1990)	"Computational Intelligence is the study of the design of intelligent agents." (Poole *et al.*, 1998)
"The study of how to make computers do things at which, at the moment, people are better." (Rich and Knight, 1991)	"AI . . . is concerned with intelligent behavior in artifacts." (Nilsson, 1998)

Figure 1.1 Some definitions of artificial intelligence, organized into four categories.

Historically, all four approaches to AI have been followed. As one might expect, a tension exists between approaches centered around humans and approaches centered around rationality.[1] A human-centered approach must be an empirical science, involving hypothesis and experimental confirmation. A rationalist approach involves a combination of mathematics and engineering. Each group has both disparaged and helped the other. Let us look at the four approaches in more detail.

Acting humanly: The Turing Test approach

TURING TEST

The **Turing Test**, proposed by Alan Turing (1950), was designed to provide a satisfactory operational definition of intelligence. Rather than proposing a long and perhaps controversial list of qualifications required for intelligence, he suggested a test based on indistinguishability from undeniably intelligent entities—human beings. The computer passes the test if a human interrogator, after posing some written questions, cannot tell whether the written responses come from a person or not. Chapter 26 discusses the details of the test and whether a computer is really intelligent if it passes. For now, we note that programming a computer to pass the test provides plenty to work on. The computer would need to possess the following capabilities:

NATURAL LANGUAGE PROCESSING

- ♢ **natural language processing** to enable it to communicate successfully in English.

[1] We should point out that, by distinguishing between *human* and *rational* behavior, we are not suggesting that humans are necessarily "irrational" in the sense of "emotionally unstable" or "insane." One merely need note that we are not perfect: we are not all chess grandmasters, even those of us who know all the rules of chess; and, unfortunately, not everyone gets an A on the exam. Some systematic errors in human reasoning are cataloged by Kahneman *et al.* (1982).

KNOWLEDGE REPRESENTATION
AUTOMATED REASONING
MACHINE LEARNING

- ♢ **knowledge representation** to store what it knows or hears;
- ♢ **automated reasoning** to use the stored information to answer questions and to draw new conclusions;
- ♢ **machine learning** to adapt to new circumstances and to detect and extrapolate patterns.

Turing's test deliberately avoided direct physical interaction between the interrogator and the computer, because *physical* simulation of a person is unnecessary for intelligence. However, the so-called **total Turing Test** includes a video signal so that the interrogator can test the subject's perceptual abilities, as well as the opportunity for the interrogator to pass physical objects "through the hatch." To pass the total Turing Test, the computer will need

TOTAL TURING TEST
COMPUTER VISION
ROBOTICS

- ♢ **computer vision** to perceive objects, and
- ♢ **robotics** to manipulate objects and move about.

These six disciplines compose most of AI, and Turing deserves credit for designing a test that remains relevant 50 years later. Yet AI researchers have devoted little effort to passing the Turing test, believing that it is more important to study the underlying principles of intelligence than to duplicate an exemplar. The quest for "artificial flight" succeeded when the Wright brothers and others stopped imitating birds and learned about aerodynamics. Aeronautical engineering texts do not define the goal of their field as making "machines that fly so exactly like pigeons that they can fool even other pigeons."

Thinking humanly: The cognitive modeling approach

If we are going to say that a given program thinks like a human, we must have some way of determining how humans think. We need to get *inside* the actual workings of human minds. There are two ways to do this: through introspection—trying to catch our own thoughts as they go by—and through psychological experiments. Once we have a sufficiently precise theory of the mind, it becomes possible to express the theory as a computer program. If the program's input/output and timing behaviors match corresponding human behaviors, that is evidence that some of the program's mechanisms could also be operating in humans. For example, Allen Newell and Herbert Simon, who developed GPS, the "General Problem Solver" (Newell and Simon, 1961), were not content to have their program solve problems correctly. They were more concerned with comparing the trace of its reasoning steps to traces of human subjects solving the same problems. The interdisciplinary field of **cognitive science** brings together computer models from AI and experimental techniques from psychology to try to construct precise and testable theories of the workings of the human mind.

COGNITIVE SCIENCE

Cognitive science is a fascinating field, worthy of an encyclopedia in itself (Wilson and Keil, 1999). We will not attempt to describe what is known of human cognition in this book. We will occasionally comment on similarities or differences between AI techniques and human cognition. Real cognitive science, however, is necessarily based on experimental investigation of actual humans or animals, and we assume that the reader has access only to a computer for experimentation.

In the early days of AI there was often confusion between the approaches: an author would argue that an algorithm performs well on a task and that it is *therefore* a good model

of human performance, or vice versa. Modern authors separate the two kinds of claims; this distinction has allowed both AI and cognitive science to develop more rapidly. The two fields continue to fertilize each other, especially in the areas of vision and natural language. Vision in particular has recently made advances via an integrated approach that considers neurophysiological evidence and computational models.

Thinking rationally: The "laws of thought" approach

SYLLOGISMS

The Greek philosopher Aristotle was one of the first to attempt to codify "right thinking," that is, irrefutable reasoning processes. His **syllogisms** provided patterns for argument structures that always yielded correct conclusions when given correct premises—for example, "Socrates is a man; all men are mortal; therefore, Socrates is mortal." These laws of thought were supposed to govern the operation of the mind; their study initiated the field called **logic**.

LOGIC

Logicians in the 19th century developed a precise notation for statements about all kinds of things in the world and about the relations among them. (Contrast this with ordinary arithmetic notation, which provides mainly for equality and inequality statements about numbers.) By 1965, programs existed that could, in principle, solve *any* solvable problem described in logical notation.[2] The so-called **logicist** tradition within artificial intelligence hopes to build on such programs to create intelligent systems.

LOGICIST

There are two main obstacles to this approach. First, it is not easy to take informal knowledge and state it in the formal terms required by logical notation, particularly when the knowledge is less than 100% certain. Second, there is a big difference between being able to solve a problem "in principle" and doing so in practice. Even problems with just a few dozen facts can exhaust the computational resources of any computer unless it has some guidance as to which reasoning steps to try first. Although both of these obstacles apply to *any* attempt to build computational reasoning systems, they appeared first in the logicist tradition.

Acting rationally: The rational agent approach

AGENT

An **agent** is just something that acts (*agent* comes from the Latin *agere*, to do). But computer agents are expected to have other attributes that distinguish them from mere "programs," such as operating under autonomous control, perceiving their environment, persisting over a prolonged time period, adapting to change, and being capable of taking on another's goals. A **rational agent** is one that acts so as to achieve the best outcome or, when there is uncertainty, the best expected outcome.

RATIONAL AGENT

In the "laws of thought" approach to AI, the emphasis was on correct inferences. Making correct inferences is sometimes *part* of being a rational agent, because one way to act rationally is to reason logically to the conclusion that a given action will achieve one's goals and then to act on that conclusion. On the other hand, correct inference is not *all* of rationality, because there are often situations where there is no provably correct thing to do, yet something must still be done. There are also ways of acting rationally that cannot be said to involve inference. For example, recoiling from a hot stove is a reflex action that is usually more successful than a slower action taken after careful deliberation.

[2] If there is no solution, the program might never stop looking for one.

All the skills needed for the Turing Test are there to allow rational actions. Thus, we need the ability to represent knowledge and reason with it because this enables us to reach good decisions in a wide variety of situations. We need to be able to generate comprehensible sentences in natural language because saying those sentences helps us get by in a complex society. We need learning not just for erudition, but because having a better idea of how the world works enables us to generate more effective strategies for dealing with it. We need visual perception not just because seeing is fun, but to get a better idea of what an action might achieve—for example, being able to see a tasty morsel helps one to move toward it.

For these reasons, the study of AI as rational-agent design has at least two advantages. First, it is more general than the "laws of thought" approach, because correct inference is just one of several possible mechanisms for achieving rationality. Second, it is more amenable to scientific development than are approaches based on human behavior or human thought because the standard of rationality is clearly defined and completely general. Human behavior, on the other hand, is well-adapted for one specific environment and is the product, in part, of a complicated and largely unknown evolutionary process that still is far from producing perfection. *This book will therefore concentrate on general principles of rational agents and on components for constructing them.* We will see that despite the apparent simplicity with which the problem can be stated, an enormous variety of issues come up when we try to solve it. Chapter 2 outlines some of these issues in more detail.

One important point to keep in mind: We will see before too long that achieving perfect rationality—always doing the right thing—is not feasible in complicated environments. The computational demands are just too high. For most of the book, however, we will adopt the working hypothesis that perfect rationality is a good starting point for analysis. It simplifies the problem and provides the appropriate setting for most of the foundational material in the field. Chapters 6 and 17 deal explicitly with the issue of **limited rationality**—acting appropriately when there is not enough time to do all the computations one might like.

LIMITED RATIONALITY

1.2 The Foundations of Artificial Intelligence

In this section, we provide a brief history of the disciplines that contributed ideas, viewpoints, and techniques to AI. Like any history, this one is forced to concentrate on a small number of people, events, and ideas and to ignore others that also were important. We organize the history around a series of questions. We certainly would not wish to give the impression that these questions are the only ones the disciplines address or that the disciplines have all been working toward AI as their ultimate fruition.

Philosophy (428 B.C.–present)

- Can formal rules be used to draw valid conclusions?
- How does the mental mind arise from a physical brain?
- Where does knowledge come from?
- How does knowledge lead to action?

Aristotle (384-322 B.C.) was the first to formulate a precise set of laws governing the rational part of the mind. He developed an informal system of syllogisms for proper reasoning, which in principle allowed one to generate conclusions mechanically, given initial premises. Much later, Ramon Lull (d. 1315) had the idea that useful reasoning could actually be carried out by a mechanical artifact. His "concept wheels" are on the cover of this book. Thomas Hobbes (1588–1679) proposed that reasoning was like numerical computation, that "we add and subtract in our silent thoughts." The automation of computation itself was already well under way; around 1500, Leonardo da Vinci (1452–1519) designed but did not build a mechanical calculator; recent reconstructions have shown the design to be functional. The first known calculating machine was constructed around 1623 by the German scientist Wilhelm Schickard (1592–1635), although the Pascaline, built in 1642 by Blaise Pascal (1623–1662), is more famous. Pascal wrote that "the arithmetical machine produces effects which appear nearer to thought than all the actions of animals." Gottfried Wilhelm Leibniz (1646–1716) built a mechanical device intended to carry out operations on concepts rather than numbers, but its scope was rather limited.

Now that we have the idea of a set of rules that can describe the formal, rational part of the mind, the next step is to consider the mind as a physical system. René Descartes (1596–1650) gave the first clear discussion of the distinction between mind and matter and of the problems that arise. One problem with a purely physical conception of the mind is that it seems to leave little room for free will: if the mind is governed entirely by physical laws, then it has no more free will than a rock "deciding" to fall toward the center of the earth. Although a strong advocate of the power of reasoning, Descartes was also a proponent of **dualism**. He held that there is a part of the human mind (or soul or spirit) that is outside of nature, exempt from physical laws. Animals, on the other hand, did not possess this dual quality; they could be treated as machines. An alternative to dualism is **materialism**, which holds that the brain's operation according to the laws of physics *constitutes* the mind. Free will is simply the way that the perception of available choices appears to the choice process.

DUALISM

MATERIALISM

Given a physical mind that manipulates knowledge, the next problem is to establish the source of knowledge. The **empiricism** movement, starting with Francis Bacon's (1561–1626) *Novum Organum*,[3] is characterized by a dictum of John Locke (1632–1704): "Nothing is in the understanding, which was not first in the senses." David Hume's (1711–1776) *A Treatise of Human Nature* (Hume, 1739) proposed what is now known as the principle of **induction**: that general rules are acquired by exposure to repeated associations between their elements. Building on the work of Ludwig Wittgenstein (1889–1951) and Bertrand Russell (1872–1970), the famous Vienna Circle, led by Rudolf Carnap (1891–1970), developed the doctrine of **logical positivism**. This doctrine holds that all knowledge can be characterized by logical theories connected, ultimately, to **observation sentences** that correspond to sensory inputs.[4] The **confirmation theory** of Carnap and Carl Hempel (1905–1997) attempted to understand how knowledge can be acquired from experience. Carnap's book *The Logical Structure of*

EMPIRICISM

INDUCTION

LOGICAL POSITIVISM

OBSERVATION SENTENCES

CONFIRMATION THEORY

[3] An update of Aristotle's *Organon*, or instrument of thought.

[4] In this picture, all meaningful statements can be verified or falsified either by analyzing the meaning of the words or by carrying out experiments. Because this rules out most of metaphysics, as was the intention, logical positivism was unpopular in some circles.

the World (1928) defined an explicit computational procedure for extracting knowledge from elementary experiences. It was probably the first theory of mind as a computational process.

The final element in the philosophical picture of the mind is the connection between knowledge and action. This question is vital to AI, because intelligence requires action as well as reasoning. Moreover, only by understanding how actions are justified can we understand how to build an agent whose actions are justifiable (or rational). Aristotle argued that actions are justified by a logical connection between goals and knowledge of the action's outcome (the last part of this extract also appears on the front cover of this book):

> But how does it happen that thinking is sometimes accompanied by action and sometimes not, sometimes by motion, and sometimes not? It looks as if almost the same thing happens as in the case of reasoning and making inferences about unchanging objects. But in that case the end is a speculative proposition . . . whereas here the conclusion which results from the two premises is an action. . . . I need covering; a cloak is a covering. I need a cloak. What I need, I have to make; I need a cloak. I have to make a cloak. And the conclusion, the "I have to make a cloak," is an action. (Nussbaum, 1978, p. 40)

In the *Nicomachean Ethics* (Book III. 3, 1112b), Aristotle further elaborates on this topic, suggesting an algorithm:

> We deliberate not about ends, but about means. For a doctor does not deliberate whether he shall heal, nor an orator whether he shall persuade, . . . They assume the end and consider how and by what means it is attained, and if it seems easily and best produced thereby; while if it is achieved by one means only they consider *how* it will be achieved by this and by what means *this* will be achieved, till they come to the first cause, . . . and what is last in the order of analysis seems to be first in the order of becoming. And if we come on an impossibility, we give up the search, e.g. if we need money and this cannot be got; but if a thing appears possible we try to do it.

Aristotle's algorithm was implemented 2300 years later by Newell and Simon in their GPS program. We would now call it a regression planning system. (See Chapter 11.)

Goal-based analysis is useful, but does not say what to do when several actions will achieve the goal, or when no action will achieve it completely. Antoine Arnauld (1612–1694) correctly described a quantitative formula for deciding what action to take in cases like this (see Chapter 16). John Stuart Mill's (1806–1873) book *Utilitarianism* (Mill, 1863) promoted the idea of rational decision criteria in all spheres of human activity. The more formal theory of decisions is discussed in the following section.

Mathematics (c. 800–present)

- What are the formal rules to draw valid conclusions?
- What can be computed?
- How do we reason with uncertain information?

Philosophers staked out most of the important ideas of AI, but the leap to a formal science required a level of mathematical formalization in three fundamental areas: logic, computation, and probability.

The idea of formal logic can be traced back to the philosophers of ancient Greece (see Chapter 7), but its mathematical development really began with the work of George Boole

(1815–1864), who worked out the details of propositional, or Boolean, logic (Boole, 1847). In 1879, Gottlob Frege (1848–1925) extended Boole's logic to include objects and relations, creating the first-order logic that is used today as the most basic knowledge representation system.[5] Alfred Tarski (1902–1983) introduced a theory of reference that shows how to relate the objects in a logic to objects in the real world. The next step was to determine the limits of what could be done with logic and computation.

ALGORITHM

The first nontrivial **algorithm** is thought to be Euclid's algorithm for computing greatest common denominators. The study of algorithms as objects in themselves goes back to al-Khowarazmi, a Persian mathematician of the 9th century, whose writings also introduced Arabic numerals and algebra to Europe. Boole and others discussed algorithms for logical deduction, and, by the late 19th century, efforts were under way to formalize general mathematical reasoning as logical deduction. In 1900, David Hilbert (1862–1943) presented a list of 23 problems that he correctly predicted would occupy mathematicians for the bulk of the century. The final problem asks whether there is an algorithm for deciding the truth of any logical proposition involving the natural numbers—the famous *Entscheidungsproblem*, or decision problem. Essentially, Hilbert was asking whether there were fundamental limits to the power of effective proof procedures. In 1930, Kurt Gödel (1906–1978) showed that there exists an effective procedure to prove any true statement in the first-order logic of Frege and Russell, but that first-order logic could not capture the principle of mathematical induction needed to characterize the natural numbers. In 1931, he showed that real limits do exist.

INCOMPLETENESS THEOREM

His **incompleteness theorem** showed that in any language expressive enough to describe the properties of the natural numbers, there are true statements that are undecidable in the sense that their truth cannot be established by any algorithm.

This fundamental result can also be interpreted as showing that there are some functions on the integers that cannot be represented by an algorithm—that is, they cannot be computed. This motivated Alan Turing (1912–1954) to try to characterize exactly which functions *are* capable of being computed. This notion is actually slightly problematic, because the notion of a computation or effective procedure really cannot be given a formal definition. However, the Church–Turing thesis, which states that the Turing machine (Turing, 1936) is capable of computing any computable function, is generally accepted as providing a sufficient definition. Turing also showed that there were some functions that no Turing machine can compute. For example, no machine can tell *in general* whether a given program will return an answer on a given input or run forever.

INTRACTABILITY

Although undecidability and noncomputability are important to an understanding of computation, the notion of **intractability** has had a much greater impact. Roughly speaking, a problem is called intractable if the time required to solve instances of the problem grows exponentially with the size of the instances. The distinction between polynomial and exponential growth in complexity was first emphasized in the mid-1960s (Cobham, 1964; Edmonds, 1965). It is important because exponential growth means that even moderately large instances cannot be solved in any reasonable time. Therefore, one should strive to divide

[5] Frege's proposed notation for first-order logic never became popular, for reasons that are apparent immediately from the example on the front cover.

the overall problem of generating intelligent behavior into tractable subproblems rather than intractable ones.

NP-COMPLETENESS

How can one recognize an intractable problem? The theory of **NP-completeness**, pioneered by Steven Cook (1971) and Richard Karp (1972), provides a method. Cook and Karp showed the existence of large classes of canonical combinatorial search and reasoning problems that are NP-complete. Any problem class to which the class of NP-complete problems can be reduced is likely to be intractable. (Although it has not been proved that NP-complete problems are necessarily intractable, most theoreticians believe it.) These results contrast with the optimism with which the popular press greeted the first computers—"Electronic Super-Brains" that were "Faster than Einstein!" Despite the increasing speed of computers, careful use of resources will characterize intelligent systems. Put crudely, the world is an *extremely* large problem instance! In recent years, AI has helped explain why some instances of NP-complete problems are hard, yet others are easy (Cheeseman *et al.*, 1991).

PROBABILITY

Besides logic and computation, the third great contribution of mathematics to AI is the theory of **probability**. The Italian Gerolamo Cardano (1501–1576) first framed the idea of probability, describing it in terms of the possible outcomes of gambling events. Probability quickly became an invaluable part of all the quantitative sciences, helping to deal with uncertain measurements and incomplete theories. Pierre Fermat (1601–1665), Blaise Pascal (1623–1662), James Bernoulli (1654–1705), Pierre Laplace (1749–1827), and others advanced the theory and introduced new statistical methods. Thomas Bayes (1702–1761) proposed a rule for updating probabilities in the light of new evidence. Bayes' rule and the resulting field called Bayesian analysis form the basis of most modern approaches to uncertain reasoning in AI systems.

Economics (1776–present)

- How should we make decisions so as to maximize payoff?
- How should we do this when others may not go along?
- How should we do this when the payoff may be far in the future?

The science of economics got its start in 1776, when Scottish philosopher Adam Smith (1723–1790) published *An Inquiry into the Nature and Causes of the Wealth of Nations*. While the ancient Greeks and others had made contributions to economic thought, Smith was the first to treat it as a science, using the idea that economies can be thought of as consisting of individual agents maximizing their own economic well-being. Most people think of economics as being about money, but economists will say that they are really studying how people make choices that lead to preferred outcomes. The mathematical treatment of "preferred outcomes" or **utility** was first formalized by Léon Walras (pronounced "Valrasse") (1834-1910) and was improved by Frank Ramsey (1931) and later by John von Neumann and Oskar Morgenstern in their book *The Theory of Games and Economic Behavior* (1944).

DECISION THEORY

Decision theory, which combines probability theory with utility theory, provides a formal and complete framework for decisions (economic or otherwise) made under uncertainty—that is, in cases where probabilistic descriptions appropriately capture the decision-maker's environment. This is suitable for "large" economies where each agent need pay no attention

to the actions of other agents as individuals. For "small" economies, the situation is much more like a **game**: the actions of one player can significantly affect the utility of another (either positively or negatively). Von Neumann and Morgenstern's development of **game theory** (see also Luce and Raiffa, 1957) included the surprising result that, for some games, a rational agent should act in a random fashion, or at least in a way that appears random to the adversaries.

GAME THEORY

For the most part, economists did not address the third question listed above, namely, how to make rational decisions when payoffs from actions are not immediate but instead result from several actions taken *in sequence*. This topic was pursued in the field of **operations research**, which emerged in World War II from efforts in Britain to optimize radar installations, and later found civilian applications in complex management decisions. The work of Richard Bellman (1957) formalized a class of sequential decision problems called **Markov decision processes**, which we study in Chapters 17 and 21.

OPERATIONS RESEARCH

Work in economics and operations research has contributed much to our notion of rational agents, yet for many years AI research developed along entirely separate paths. One reason was the apparent **complexity** of making rational decisions. Herbert Simon (1916–2001), the pioneering AI researcher, won the Nobel prize in economics in 1978 for his early work showing that models based on **satisficing**–making decisions that are "good enough," rather than laboriously calculating an optimal decision–gave a better description of actual human behavior (Simon, 1947). In the 1990s, there has been a resurgence of interest in decision-theoretic techniques for agent systems (Wellman, 1995).

SATISFICING

Neuroscience (1861–present)

- How do brains process information?

NEUROSCIENCE

Neuroscience is the study of the nervous system, particularly the brain. The exact way in which the brain enables thought is one of the great mysteries of science. It has been appreciated for thousands of years that the brain is somehow involved in thought, because of the evidence that strong blows to the head can lead to mental incapacitation. It has also long been known that human brains are somehow different; in about 335 B.C. Aristotle wrote, "Of all the animals, man has the largest brain in proportion to his size." [6] Still, it was not until the middle of the 18th century that the brain was widely recognized as the seat of consciousness. Before then, candidate locations included the heart, the spleen, and the pineal gland.

Paul Broca's (1824–1880) study of aphasia (speech deficit) in brain-damaged patients in 1861 reinvigorated the field and persuaded the medical establishment of the existence of localized areas of the brain responsible for specific cognitive functions. In particular, he showed that speech production was localized to a portion of the left hemisphere now called Broca's area.[7] By that time, it was known that the brain consisted of nerve cells or **neurons**, but it was not until 1873 that Camillo Golgi (1843–1926) developed a staining technique allowing the observation of individual neurons in the brain (see Figure 1.2). This technique

NEURONS

[6] Since then, it has been discovered that some species of dolphins and whales have relatively larger brains. The large size of human brains is now thought to be enabled in part by recent improvements in its cooling system.

[7] Many cite Alexander Hood (1824) as a possible prior source.

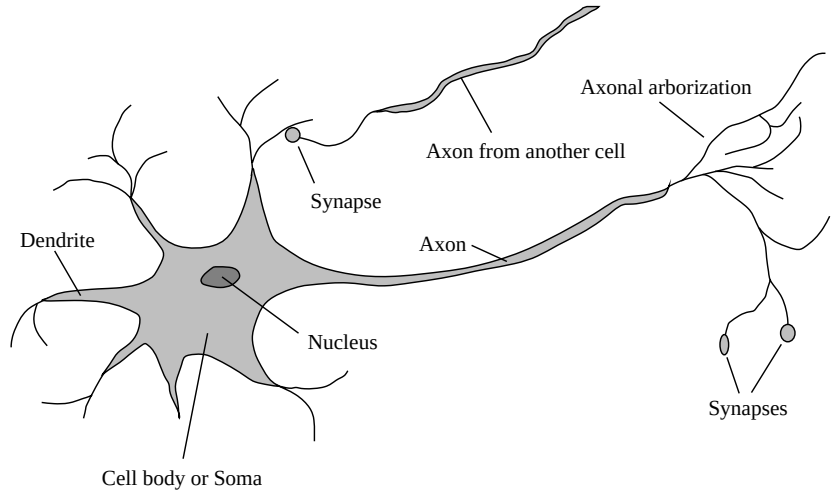

Figure 1.2 The parts of a nerve cell or neuron. Each neuron consists of a cell body, or soma, that contains a cell nucleus. Branching out from the cell body are a number of fibers called dendrites and a single long fiber called the axon. The axon stretches out for a long distance, much longer than the scale in this diagram indicates. Typically they are 1 cm long (100 times the diameter of the cell body), but can reach up to 1 meter. A neuron makes connections with 10 to 100,000 other neurons at junctions called synapses. Signals are propagated from neuron to neuron by a complicated electrochemical reaction. The signals control brain activity in the short term, and also enable long-term changes in the position and connectivity of neurons. These mechanisms are thought to form the basis for learning in the brain. Most information processing goes on in the cerebral cortex, the outer layer of the brain. The basic organizational unit appears to be a column of tissue about 0.5 mm in diameter, extending the full depth of the cortex, which is about 4 mm in humans. A column contains about 20,000 neurons.

was used by Santiago Ramon y Cajal (1852–1934) in his pioneering studies of the brain's neuronal structures.[8]

We now have some data on the mapping between areas of the brain and the parts of the body that they control or from which they receive sensory input. Such mappings are able to change radically over the course of a few weeks, and some animals seem to have multiple maps. Moreover, we do not fully understand how other areas can take over functions when one area is damaged. There is almost no theory on how an individual memory is stored.

The measurement of intact brain activity began in 1929 with the invention by Hans Berger of the electroencephalograph (EEG). The recent development of functional magnetic resonance imaging (fMRI) (Ogawa *et al.*, 1990) is giving neuroscientists unprecedentedly detailed images of brain activity, enabling measurements that correspond in interesting ways to ongoing cognitive processes. These are augmented by advances in single-cell recording of

[8] Golgi persisted in his belief that the brain's functions were carried out primarily in a continuous medium in which neurons were embedded, whereas Cajal propounded the "neuronal doctrine." The two shared the Nobel prize in 1906 but gave rather antagonistic acceptance speeches.

	Computer	Human Brain
Computational units	1 CPU, 10^8 gates	10^{11} neurons
Storage units	10^{10} bits RAM	10^{11} neurons
	10^{11} bits disk	10^{14} synapses
Cycle time	10^{-9} sec	10^{-3} sec
Bandwidth	10^{10} bits/sec	10^{14} bits/sec
Memory updates/sec	10^9	10^{14}

Figure 1.3 A crude comparison of the raw computational resources available to computers (*circa* 2003) and brains. The computer's numbers have all increased by at least a factor of 10 since the first edition of this book, and are expected to do so again this decade. The brain's numbers have not changed in the last 10,000 years.

neuron activity. Despite these advances, we are still a long way from understanding how any of these cognitive processes actually work.

The truly amazing conclusion is that *a collection of simple cells can lead to thought, action, and consciousness* or, in other words, that *brains cause minds* (Searle, 1992). The only real alternative theory is mysticism: that there is some mystical realm in which minds operate that is beyond physical science.

Brains and digital computers perform quite different tasks and have different properties. Figure 1.3 shows that there are 1000 times more neurons in the typical human brain than there are gates in the CPU of a typical high-end computer. Moore's Law[9] predicts that the CPU's gate count will equal the brain's neuron count around 2020. Of course, little can be inferred from such predictions; moreover, the difference in storage capacity is minor compared to the difference in switching speed and in parallelism. Computer chips can execute an instruction in a nanosecond, whereas neurons are millions of times slower. Brains more than make up for this, however, because all the neurons and synapses are active simultaneously, whereas most current computers have only one or at most a few CPUs. Thus, *even though a computer is a million times faster in raw switching speed, the brain ends up being 100,000 times faster at what it does.*

Psychology (1879–present)

- How do humans and animals think and act?

The origins of scientific psychology are usually traced to the work of the German physicist Hermann von Helmholtz (1821–1894) and his student Wilhelm Wundt (1832–1920). Helmholtz applied the scientific method to the study of human vision, and his *Handbook of Physiological Optics* is even now described as "the single most important treatise on the physics and physiology of human vision" (Nalwa, 1993, p.15). In 1879, Wundt opened the first laboratory of experimental psychology at the University of Leipzig. Wundt insisted on carefully controlled experiments in which his workers would perform a perceptual or associa-

[9] Moore's Law says that the number of transistors per square inch doubles every 1 to 1.5 years. Human brain capacity doubles roughly every 2 to 4 million years.

tive task while introspecting on their thought processes. The careful controls went a long way toward making psychology a science, but the subjective nature of the data made it unlikely that an experimenter would ever disconfirm his or her own theories. Biologists studying animal behavior, on the other hand, lacked introspective data and developed an objective methodology, as described by H. S. Jennings (1906) in his influential work *Behavior of the Lower Organisms*. Applying this viewpoint to humans, the **behaviorism** movement, led by John Watson (1878–1958), rejected *any* theory involving mental processes on the grounds that introspection could not provide reliable evidence. Behaviorists insisted on studying only objective measures of the percepts (or *stimulus*) given to an animal and its resulting actions (or *response*). Mental constructs such as knowledge, beliefs, goals, and reasoning steps were dismissed as unscientific "folk psychology." Behaviorism discovered a lot about rats and pigeons, but had less success at understanding humans. Nevertheless, it exerted a strong hold on psychology (especially in the United States) from about 1920 to 1960.

BEHAVIORISM

The view of the brain as an information-processing device, which is a principal characteristic of **cognitive psychology**, can be traced back at least to the works of William James[10] (1842–1910). Helmholtz also insisted that perception involved a form of unconscious logical inference. The cognitive viewpoint was largely eclipsed by behaviorism in the United States, but at Cambridge's Applied Psychology Unit, directed by Frederic Bartlett (1886–1969), cognitive modeling was able to flourish. *The Nature of Explanation*, by Bartlett's student and successor Kenneth Craik (1943), forcefully reestablished the legitimacy of such "mental" terms as beliefs and goals, arguing that they are just as scientific as, say, using pressure and temperature to talk about gases, despite their being made of molecules that have neither. Craik specified the three key steps of a knowledge-based agent: (1) the stimulus must be translated into an internal representation, (2) the representation is manipulated by cognitive processes to derive new internal representations, and (3) these are in turn retranslated back into action. He clearly explained why this was a good design for an agent:

COGNITIVE PSYCHOLOGY

> If the organism carries a "small-scale model" of external reality and of its own possible actions within its head, it is able to try out various alternatives, conclude which is the best of them, react to future situations before they arise, utilize the knowledge of past events in dealing with the present and future, and in every way to react in a much fuller, safer, and more competent manner to the emergencies which face it. (Craik, 1943)

After Craik's death in a bicycle accident in 1945, his work was continued by Donald Broadbent, whose book *Perception and Communication* (1958) included some of the first information-processing models of psychological phenomena. Meanwhile, in the United States, the development of computer modeling led to the creation of the field of **cognitive science**. The field can be said to have started at a workshop in September 1956 at MIT. (We shall see that this is just two months after the conference at which AI itself was "born.") At the workshop, George Miller presented *The Magic Number Seven*, Noam Chomsky presented *Three Models of Language*, and Allen Newell and Herbert Simon presented *The Logic Theory Machine*. These three influential papers showed how computer models could be used to

COGNITIVE SCIENCE

[10] William James was the brother of novelist Henry James. It is said that Henry wrote fiction as if it were psychology and William wrote psychology as if it were fiction.

address the psychology of memory, language, and logical thinking, respectively. It is now a common view among psychologists that "a cognitive theory should be like a computer program" (Anderson, 1980), that is, it should describe a detailed information-processing mechanism whereby some cognitive function might be implemented.

Computer engineering (1940–present)

- How can we build an efficient computer?

For artificial intelligence to succeed, we need two things: intelligence and an artifact. The computer has been the artifact of choice. The modern digital electronic computer was invented independently and almost simultaneously by scientists in three countries embattled in World War II. The first *operational* computer was the electromechanical Heath Robinson,[11] built in 1940 by Alan Turing's team for a single purpose: deciphering German messages. In 1943, the same group developed the Colossus, a powerful general-purpose machine based on vacuum tubes.[12] The first operational *programmable* computer was the Z-3, the invention of Konrad Zuse in Germany in 1941. Zuse also invented floating-point numbers and the first high-level programming language, Plankalkül. The first *electronic* computer, the ABC, was assembled by John Atanasoff and his student Clifford Berry between 1940 and 1942 at Iowa State University. Atanasoff's research received little support or recognition; it was the ENIAC, developed as part of a secret military project at the University of Pennsylvania by a team including John Mauchly and John Eckert, that proved to be the most influential forerunner of modern computers.

In the half-century since then, each generation of computer hardware has brought an increase in speed and capacity and a decrease in price. Performance doubles every 18 months or so, with a decade or two to go at this rate of increase. After that, we will need molecular engineering or some other new technology.

Of course, there were calculating devices before the electronic computer. The earliest automated machines, dating from the 17th century, were discussed on page 6. The first *programmable* machine was a loom devised in 1805 by Joseph Marie Jacquard (1752–1834) that used punched cards to store instructions for the pattern to be woven. In the mid-19th century, Charles Babbage (1792–1871) designed two machines, neither of which he completed. The "Difference Engine," which appears on the cover of this book, was intended to compute mathematical tables for engineering and scientific projects. It was finally built and shown to work in 1991 at the Science Museum in London (Swade, 1993). Babbage's "Analytical Engine" was far more ambitious: it included addressable memory, stored programs, and conditional jumps and was the first artifact capable of universal computation. Babbage's colleague Ada Lovelace, daughter of the poet Lord Byron, was perhaps the world's first programmer. (The programming language Ada is named after her.) She wrote programs for the unfinished Analytical Engine and even speculated that the machine could play chess or compose music.

[11] Heath Robinson was a cartoonist famous for his depictions of whimsical and absurdly complicated contraptions for everyday tasks such as buttering toast.

[12] In the postwar period, Turing wanted to use these computers for AI research—for example, one of the first chess programs (Turing *et al.*, 1953). His efforts were blocked by the British government.

AI also owes a debt to the software side of computer science, which has supplied the operating systems, programming languages, and tools needed to write modern programs (and papers about them). But this is one area where the debt has been repaid: work in AI has pioneered many ideas that have made their way back to mainstream computer science, including time sharing, interactive interpreters, personal computers with windows and mice, rapid development environments, the linked list data type, automatic storage management, and key concepts of symbolic, functional, dynamic, and object-oriented programming.

Control theory and Cybernetics (1948–present)

- How can artifacts operate under their own control?

Ktesibios of Alexandria (c. 250 B.C.) built the first self-controlling machine: a water clock with a regulator that kept the flow of water running through it at a constant, predictable pace. This invention changed the definition of what an artifact could do. Previously, only living things could modify their behavior in response to changes in the environment. Other examples of self-regulating feedback control systems include the steam engine governor, created by James Watt (1736–1819), and the thermostat, invented by Cornelis Drebbel (1572–1633), who also invented the submarine. The mathematical theory of stable feedback systems was developed in the 19th century.

CONTROL THEORY

The central figure in the creation of what is now called **control theory** was Norbert Wiener (1894–1964). Wiener was a brilliant mathematician who worked with Bertrand Russell, among others, before developing an interest in biological and mechanical control systems and their connection to cognition. Like Craik (who also used control systems as psychological models), Wiener and his colleagues Arturo Rosenblueth and Julian Bigelow challenged the behaviorist orthodoxy (Rosenblueth *et al.*, 1943). They viewed purposive behavior as arising from a regulatory mechanism trying to minimize "error"—the difference between current state and goal state. In the late 1940s, Wiener, along with Warren McCulloch, Walter Pitts, and John von Neumann, organized a series of conferences that explored the new mathematical and computational models of cognition and influenced many other researchers in the behavioral sciences. Wiener's book *Cybernetics* (1948) became a bestseller and awoke the public to the possibility of artificially intelligent machines.

CYBERNETICS

OBJECTIVE FUNCTION

Modern control theory, especially the branch known as stochastic optimal control, has as its goal the design of systems that maximize an **objective function** over time. This roughly matches our view of AI: designing systems that behave optimally. Why, then, are AI and control theory two different fields, especially given the close connections among their founders? The answer lies in the close coupling between the mathematical techniques that were familiar to the participants and the corresponding sets of problems that were encompassed in each world view. Calculus and matrix algebra, the tools of control theory, lend themselves to systems that are describable by fixed sets of continuous variables; furthermore, exact analysis is typically feasible only for *linear* systems. AI was founded in part as a way to escape from the limitations of the mathematics of control theory in the 1950s. The tools of logical inference and computation allowed AI researchers to consider some problems such as language, vision, and planning, that fell completely outside the control theorist's purview.

Linguistics (1957–present)

- How does language relate to thought?

In 1957, B. F. Skinner published *Verbal Behavior*. This was a comprehensive, detailed account of the behaviorist approach to language learning, written by the foremost expert in the field. But curiously, a review of the book became as well known as the book itself, and served to almost kill off interest in behaviorism. The author of the review was Noam Chomsky, who had just published a book on his own theory, *Syntactic Structures*. Chomsky showed how the behaviorist theory did not address the notion of creativity in language—it did not explain how a child could understand and make up sentences that he or she had never heard before. Chomsky's theory—based on syntactic models going back to the Indian linguist Panini (c. 350 B.C.)—could explain this, and unlike previous theories, it was formal enough that it could in principle be programmed.

COMPUTATIONAL LINGUISTICS

Modern linguistics and AI, then, were "born" at about the same time, and grew up together, intersecting in a hybrid field called **computational linguistics** or **natural language processing**. The problem of understanding language soon turned out to be considerably more complex than it seemed in 1957. Understanding language requires an understanding of the subject matter and context, not just an understanding of the structure of sentences. This might seem obvious, but it was not widely appreciated until the 1960s. Much of the early work in **knowledge representation** (the study of how to put knowledge into a form that a computer can reason with) was tied to language and informed by research in linguistics, which was connected in turn to decades of work on the philosophical analysis of language.

1.3 The History of Artificial Intelligence

With the background material behind us, we are ready to cover the development of AI itself.

The gestation of artificial intelligence (1943–1955)

The first work that is now generally recognized as AI was done by Warren McCulloch and Walter Pitts (1943). They drew on three sources: knowledge of the basic physiology and function of neurons in the brain; a formal analysis of propositional logic due to Russell and Whitehead; and Turing's theory of computation. They proposed a model of artificial neurons in which each neuron is characterized as being "on" or "off," with a switch to "on" occurring in response to stimulation by a sufficient number of neighboring neurons. The state of a neuron was conceived of as "factually equivalent to a proposition which proposed its adequate stimulus." They showed, for example, that any computable function could be computed by some network of connected neurons, and that all the logical connectives (and, or, not, etc.) could be implemented by simple net structures. McCulloch and Pitts also suggested that suitably defined networks could learn. Donald Hebb (1949) demonstrated a simple updating rule for modifying the connection strengths between neurons. His rule, now called **Hebbian learning**, remains an influential model to this day.

Two graduate students in the Princeton mathematics department, Marvin Minsky and Dean Edmonds, built the first neural network computer in 1951. The SNARC, as it was called, used 3000 vacuum tubes and a surplus automatic pilot mechanism from a B-24 bomber to simulate a network of 40 neurons. Minsky's Ph.D. committee was skeptical about whether this kind of work should be considered mathematics, but von Neumann reportedly said, "If it isn't now, it will be someday." Minsky was later to prove influential theorems showing the limitations of neural network research.

There were a number of early examples of work that can be characterized as AI, but it was Alan Turing who first articulated a complete vision of AI in his 1950 article "Computing Machinery and Intelligence." Therein, he introduced the Turing test, machine learning, genetic algorithms, and reinforcement learning.

The birth of artificial intelligence (1956)

Princeton was home to another influential figure in AI, John McCarthy. After graduation, McCarthy moved to Dartmouth College, which was to become the official birthplace of the field. McCarthy convinced Minsky, Claude Shannon, and Nathaniel Rochester to help him bring together U.S. researchers interested in automata theory, neural nets, and the study of intelligence. They organized a two-month workshop at Dartmouth in the summer of 1956. There were 10 attendees in all, including Trenchard More from Princeton, Arthur Samuel from IBM, and Ray Solomonoff and Oliver Selfridge from MIT.

Two researchers from Carnegie Tech,[13] Allen Newell and Herbert Simon, rather stole the show. Although the others had ideas and in some cases programs for particular applications such as checkers, Newell and Simon already had a reasoning program, the Logic Theorist (LT), about which Simon claimed, "We have invented a computer program capable of thinking non-numerically, and thereby solved the venerable mind–body problem."[14] Soon after the workshop, the program was able to prove most of the theorems in Chapter 2 of Russell and Whitehead's *Principia Mathematica*. Russell was reportedly delighted when Simon showed him that the program had come up with a proof for one theorem that was shorter than the one in *Principia*. The editors of the *Journal of Symbolic Logic* were less impressed; they rejected a paper coauthored by Newell, Simon, and Logic Theorist.

The Dartmouth workshop did not lead to any new breakthroughs, but it did introduce all the major figures to each other. For the next 20 years, the field would be dominated by these people and their students and colleagues at MIT, CMU, Stanford, and IBM. Perhaps the longest-lasting thing to come out of the workshop was an agreement to adopt McCarthy's new name for the field: **artificial intelligence**. Perhaps "computational rationality" would have been better, but "AI" has stuck.

Looking at the proposal for the Dartmouth workshop (McCarthy *et al.*, 1955), we can see why it was necessary for AI to become a separate field. Why couldn't all the work done

[13] Now Carnegie Mellon University (CMU).

[14] Newell and Simon also invented a list-processing language, IPL, to write LT. They had no compiler, and translated it into machine code by hand. To avoid errors, they worked in parallel, calling out binary numbers to each other as they wrote each instruction to make sure they agreed.

in AI have taken place under the name of control theory, or operations research, or decision theory, which, after all, have objectives similar to those of AI? Or why isn't AI a branch of mathematics? The first answer is that AI from the start embraced the idea of duplicating human faculties like creativity, self-improvement, and language use. None of the other fields were addressing these issues. The second answer is methodology. AI is the only one of these fields that is clearly a branch of computer science (although operations research does share an emphasis on computer simulations), and AI is the only field to attempt to build machines that will function autonomously in complex, changing environments.

Early enthusiasm, great expectations (1952–1969)

The early years of AI were full of successes—in a limited way. Given the primitive computers and programming tools of the time, and the fact that only a few years earlier computers were seen as things that could do arithmetic and no more, it was astonishing whenever a computer did anything remotely clever. The intellectual establishment, by and large, preferred to believe that "a machine can never do X." (See Chapter 26 for a long list of X's gathered by Turing.) AI researchers naturally responded by demonstrating one X after another. John McCarthy referred to this period as the "Look, Ma, no hands!" era.

Newell and Simon's early success was followed up with the General Problem Solver, or GPS. Unlike Logic Theorist, this program was designed from the start to imitate human problem-solving protocols. Within the limited class of puzzles it could handle, it turned out that the order in which the program considered subgoals and possible actions was similar to that in which humans approached the same problems. Thus, GPS was probably the first program to embody the "thinking humanly" approach. The success of GPS and subsequent programs as models of cognition led Newell and Simon (1976) to formulate the famous **physical symbol system** hypothesis, which states that "a physical symbol system has the necessary and sufficient means for general intelligent action." What they meant is that any system (human or machine) exhibiting intelligence must operate by manipulating data structures composed of symbols. We will see later that this hypothesis has been challenged from many directions.

PHYSICAL SYMBOL SYSTEM

At IBM, Nathaniel Rochester and his colleagues produced some of the first AI programs. Herbert Gelernter (1959) constructed the Geometry Theorem Prover, which was able to prove theorems that many students of mathematics would find quite tricky. Starting in 1952, Arthur Samuel wrote a series of programs for checkers (draughts) that eventually learned to play at a strong amateur level. Along the way, he disproved the idea that computers can do only what they are told to: his program quickly learned to play a better game than its creator. The program was demonstrated on television in February 1956, creating a very strong impression. Like Turing, Samuel had trouble finding computer time. Working at night, he used machines that were still on the testing floor at IBM's manufacturing plant. Chapter 6 covers game playing, and Chapter 21 describes and expands on the learning techniques used by Samuel.

John McCarthy moved from Dartmouth to MIT and there made three crucial contributions in one historic year: 1958. In MIT AI Lab Memo No. 1, McCarthy defined the high-level language **Lisp**, which was to become the dominant AI programming language. Lisp is the

LISP

second-oldest major high-level language in current use, one year younger than FORTRAN. With Lisp, McCarthy had the tool he needed, but access to scarce and expensive computing resources was also a serious problem. In response, he and others at MIT invented time sharing. Also in 1958, McCarthy published a paper entitled *Programs with Common Sense*, in which he described the Advice Taker, a hypothetical program that can be seen as the first complete AI system. Like the Logic Theorist and Geometry Theorem Prover, McCarthy's program was designed to use knowledge to search for solutions to problems. But unlike the others, it was to embody general knowledge of the world. For example, he showed how some simple axioms would enable the program to generate a plan to drive to the airport to catch a plane. The program was also designed so that it could accept new axioms in the normal course of operation, thereby allowing it to achieve competence in new areas *without being reprogrammed*. The Advice Taker thus embodied the central principles of knowledge representation and reasoning: that it is useful to have a formal, explicit representation of the world and of the way an agent's actions affect the world and to be able to manipulate these representations with deductive processes. It is remarkable how much of the 1958 paper remains relevant even today.

1958 also marked the year that Marvin Minsky moved to MIT. His initial collaboration with McCarthy did not last, however. McCarthy stressed representation and reasoning in formal logic, whereas Minsky was more interested in getting programs to work and eventually developed an anti-logical outlook. In 1963, McCarthy started the AI lab at Stanford. His plan to use logic to build the ultimate Advice Taker was advanced by J. A. Robinson's discovery of the resolution method (a complete theorem-proving algorithm for first-order logic; see Chapter 9). Work at Stanford emphasized general-purpose methods for logical reasoning. Applications of logic included Cordell Green's question-answering and planning systems (Green, 1969b) and the Shakey robotics project at the new Stanford Research Institute (SRI). The latter project, discussed further in Chapter 25, was the first to demonstrate the complete integration of logical reasoning and physical activity.

MICROWORLDS

Minsky supervised a series of students who chose limited problems that appeared to require intelligence to solve. These limited domains became known as **microworlds**. James Slagle's SAINT program (1963a) was able to solve closed-form calculus integration problems typical of first-year college courses. Tom Evans's ANALOGY program (1968) solved geometric analogy problems that appear in IQ tests, such as the one in Figure 1.4. Daniel Bobrow's STUDENT program (1967) solved algebra story problems, such as the following:

> If the number of customers Tom gets is twice the square of 20 percent of the number of advertisements he runs, and the number of advertisements he runs is 45, what is the number of customers Tom gets?

The most famous microworld was the blocks world, which consists of a set of solid blocks placed on a tabletop (or more often, a simulation of a tabletop), as shown in Figure 1.5. A typical task in this world is to rearrange the blocks in a certain way, using a robot hand that can pick up one block at a time. The blocks world was home to the vision project of David Huffman (1971), the vision and constraint-propagation work of David Waltz (1975), the learning theory of Patrick Winston (1970), the natural language understanding program

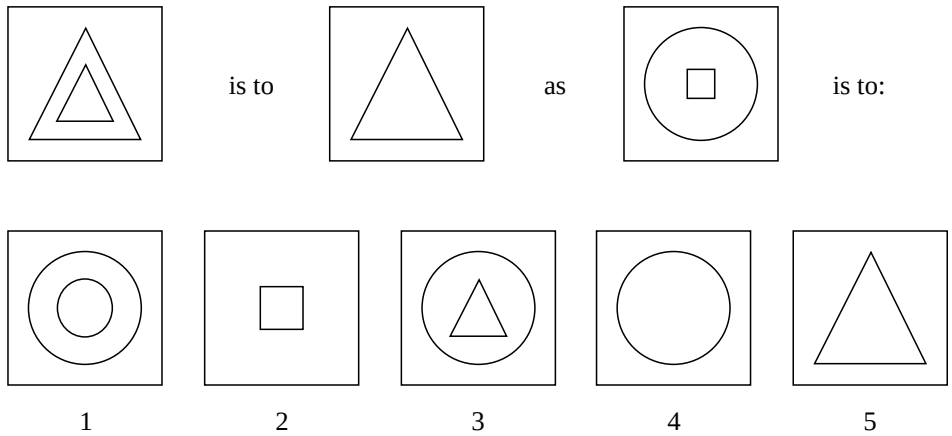

Figure 1.4 An example problem solved by Evans's ANALOGY program.

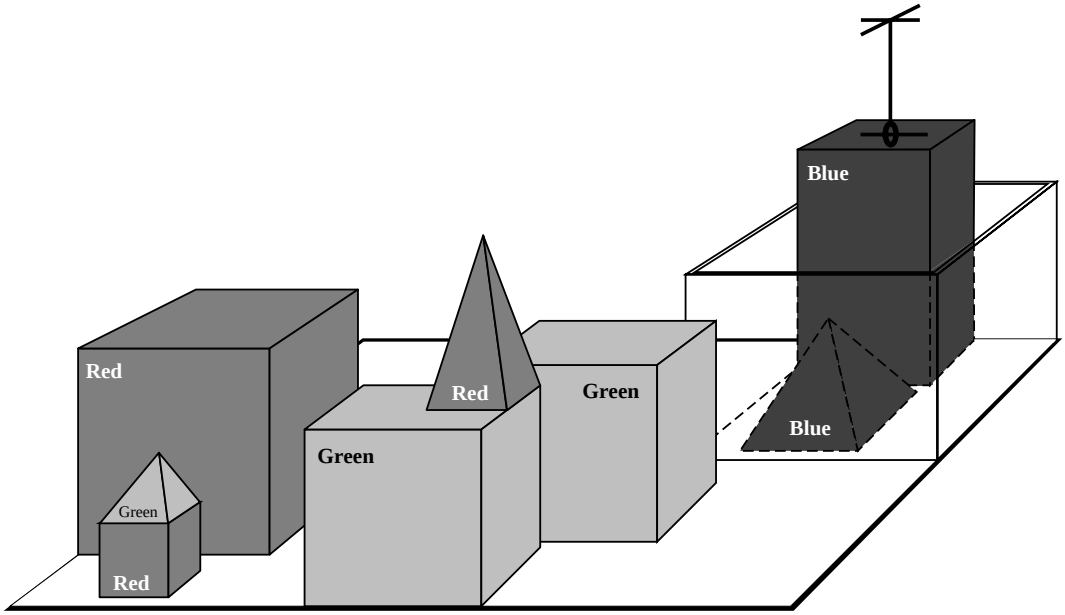

Figure 1.5 A scene from the blocks world. SHRDLU (Winograd, 1972) has just completed the command, "Find a block which is taller than the one you are holding and put it in the box."

of Terry Winograd (1972), and the planner of Scott Fahlman (1974).

Early work building on the neural networks of McCulloch and Pitts also flourished. The work of Winograd and Cowan (1963) showed how a large number of elements could collectively represent an individual concept, with a corresponding increase in robustness and parallelism. Hebb's learning methods were enhanced by Bernie Widrow (Widrow and Hoff,

1960; Widrow, 1962), who called his networks **adalines**, and by Frank Rosenblatt (1962) with his **perceptrons**. Rosenblatt proved the **perceptron convergence theorem**, showing that his learning algorithm could adjust the connection strengths of a perceptron to match any input data, provided such a match existed. These topics are covered in Chapter 20.

A dose of reality (1966–1973)

From the beginning, AI researchers were not shy about making predictions of their coming successes. The following statement by Herbert Simon in 1957 is often quoted:

> It is not my aim to surprise or shock you—but the simplest way I can summarize is to say that there are now in the world machines that think, that learn and that create. Moreover, their ability to do these things is going to increase rapidly until—in a visible future—the range of problems they can handle will be coextensive with the range to which the human mind has been applied.

Terms such as "visible future" can be interpreted in various ways, but Simon also made a more concrete prediction: that within 10 years a computer would be chess champion, and a significant mathematical theorem would be proved by machine. These predictions came true (or approximately true) within 40 years rather than 10. Simon's over-confidence was due to the promising performance of early AI systems on simple examples. In almost all cases, however, these early systems turned out to fail miserably when tried out on wider selections of problems and on more difficult problems.

The first kind of difficulty arose because most early programs contained little or no knowledge of their subject matter; they succeeded by means of simple syntactic manipulations. A typical story occurred in early machine translation efforts, which were generously funded by the U.S. National Research Council in an attempt to speed up the translation of Russian scientific papers in the wake of the Sputnik launch in 1957. It was thought initially that simple syntactic transformations based on the grammars of Russian and English, and word replacement using an electronic dictionary, would suffice to preserve the exact meanings of sentences. The fact is that translation requires general knowledge of the subject matter in order to resolve ambiguity and establish the content of the sentence. The famous re-translation of "the spirit is willing but the flesh is weak" as "the vodka is good but the meat is rotten" illustrates the difficulties encountered. In 1966, a report by an advisory committee found that "there has been no machine translation of general scientific text, and none is in immediate prospect." All U.S. government funding for academic translation projects was canceled. Today, machine translation is an imperfect but widely used tool for technical, commercial, government, and Internet documents.

The second kind of difficulty was the intractability of many of the problems that AI was attempting to solve. Most of the early AI programs solved problems by trying out different combinations of steps until the solution was found. This strategy worked initially because microworlds contained very few objects and hence very few possible actions and very short solution sequences. Before the theory of computational complexity was developed, it was widely thought that "scaling up" to larger problems was simply a matter of faster hardware and larger memories. The optimism that accompanied the development of resolution theorem

proving, for example, was soon dampened when researchers failed to prove theorems involving more than a few dozen facts. *The fact that a program can find a solution in principle does not mean that the program contains any of the mechanisms needed to find it in practice.*

MACHINE EVOLUTION

The illusion of unlimited computational power was not confined to problem-solving programs. Early experiments in **machine evolution** (now called **genetic algorithms**) (Friedberg, 1958; Friedberg *et al.*, 1959) were based on the undoubtedly correct belief that by making an appropriate series of small mutations to a machine code program, one can generate a program with good performance for any particular simple task. The idea, then, was to try random mutations with a selection process to preserve mutations that seemed useful. Despite thousands of hours of CPU time, almost no progress was demonstrated. Modern genetic algorithms use better representations and have shown more success.

Failure to come to grips with the "combinatorial explosion" was one of the main criticisms of AI contained in the Lighthill report (Lighthill, 1973), which formed the basis for the decision by the British government to end support for AI research in all but two universities. (Oral tradition paints a somewhat different and more colorful picture, with political ambitions and personal animosities whose description is beside the point.)

A third difficulty arose because of some fundamental limitations on the basic structures being used to generate intelligent behavior. For example, Minsky and Papert's book *Perceptrons* (1969) proved that, although perceptrons (a simple form of neural network) could be shown to learn anything they were capable of representing, they could represent very little. In particular, a two-input perceptron could not be trained to recognize when its two inputs were different. Although their results did not apply to more complex, multilayer networks, research funding for neural-net research soon dwindled to almost nothing. Ironically, the new back-propagation learning algorithms for multilayer networks that were to cause an enormous resurgence in neural-net research in the late 1980s were actually discovered first in 1969 (Bryson and Ho, 1969).

Knowledge-based systems: The key to power? (1969–1979)

WEAK METHODS

The picture of problem solving that had arisen during the first decade of AI research was of a general-purpose search mechanism trying to string together elementary reasoning steps to find complete solutions. Such approaches have been called **weak methods**, because, although general, they do not scale up to large or difficult problem instances. The alternative to weak methods is to use more powerful, domain-specific knowledge that allows larger reasoning steps and can more easily handle typically occurring cases in narrow areas of expertise. One might say that to solve a hard problem, you have to almost know the answer already.

The DENDRAL program (Buchanan *et al.*, 1969) was an early example of this approach. It was developed at Stanford, where Ed Feigenbaum (a former student of Herbert Simon), Bruce Buchanan (a philosopher turned computer scientist), and Joshua Lederberg (a Nobel laureate geneticist) teamed up to solve the problem of inferring molecular structure from the information provided by a mass spectrometer. The input to the program consists of the elementary formula of the molecule (e.g., $C_6H_{13}NO_2$) and the mass spectrum giving the masses of the various fragments of the molecule generated when it is bombarded by an electron beam.

For example, the mass spectrum might contain a peak at $m = 15$, corresponding to the mass of a methyl (CH_3) fragment.

The naive version of the program generated all possible structures consistent with the formula, and then predicted what mass spectrum would be observed for each, comparing this with the actual spectrum. As one might expect, this is intractable for decent-sized molecules. The DENDRAL researchers consulted analytical chemists and found that they worked by looking for well-known patterns of peaks in the spectrum that suggested common substructures in the molecule. For example, the following rule is used to recognize a ketone (C=O) subgroup (which weighs 28):

> **if** there are two peaks at x_1 and x_2 such that
> (a) $x_1 + x_2 = M + 28$ (M is the mass of the whole molecule);
> (b) $x_1 - 28$ is a high peak;
> (c) $x_2 - 28$ is a high peak;
> (d) At least one of x_1 and x_2 is high.
> **then** there is a ketone subgroup

Recognizing that the molecule contains a particular substructure reduces the number of possible candidates enormously. DENDRAL was powerful because

> All the relevant theoretical knowledge to solve these problems has been mapped over from its general form in the [spectrum prediction component] ("first principles") to efficient special forms ("cookbook recipes"). (Feigenbaum *et al.*, 1971)

The significance of DENDRAL was that it was the first successful *knowledge-intensive* system: its expertise derived from large numbers of special-purpose rules. Later systems also incorporated the main theme of McCarthy's Advice Taker approach—the clean separation of the knowledge (in the form of rules) from the reasoning component.

With this lesson in mind, Feigenbaum and others at Stanford began the Heuristic Programming Project (HPP), to investigate the extent to which the new methodology of **expert systems** could be applied to other areas of human expertise. The next major effort was in the area of medical diagnosis. Feigenbaum, Buchanan, and Dr. Edward Shortliffe developed MYCIN to diagnose blood infections. With about 450 rules, MYCIN was able to perform as well as some experts, and considerably better than junior doctors. It also contained two major differences from DENDRAL. First, unlike the DENDRAL rules, no general theoretical model existed from which the MYCIN rules could be deduced. They had to be acquired from extensive interviewing of experts, who in turn acquired them from textbooks, other experts, and direct experience of cases. Second, the rules had to reflect the uncertainty associated with medical knowledge. MYCIN incorporated a calculus of uncertainty called **certainty factors** (see Chapter 13), which seemed (at the time) to fit well with how doctors assessed the impact of evidence on the diagnosis.

EXPERT SYSTEMS

The importance of domain knowledge was also apparent in the area of understanding natural language. Although Winograd's SHRDLU system for understanding natural language had engendered a good deal of excitement, its dependence on syntactic analysis caused some of the same problems as occurred in the early machine translation work. It was able to overcome ambiguity and understand pronoun references, but this was mainly because it was

designed specifically for one area—the blocks world. Several researchers, including Eugene Charniak, a fellow graduate student of Winograd's at MIT, suggested that robust language understanding would require general knowledge about the world and a general method for using that knowledge.

At Yale, the linguist-turned-AI-researcher Roger Schank emphasized this point, claiming, "There is no such thing as syntax," which upset a lot of linguists, but did serve to start a useful discussion. Schank and his students built a series of programs (Schank and Abelson, 1977; Wilensky, 1978; Schank and Riesbeck, 1981; Dyer, 1983) that all had the task of understanding natural language. The emphasis, however, was less on language *per se* and more on the problems of representing and reasoning with the knowledge required for language understanding. The problems included representing stereotypical situations (Cullingford, 1981), describing human memory organization (Rieger, 1976; Kolodner, 1983), and understanding plans and goals (Wilensky, 1983).

The widespread growth of applications to real-world problems caused a concurrent increase in the demands for workable knowledge representation schemes. A large number of different representation and reasoning languages were developed. Some were based on logic—for example, the Prolog language became popular in Europe, and the PLANNER family in the United States. Others, following Minsky's idea of **frames** (1975), adopted a more structured approach, assembling facts about particular object and event types and arranging the types into a large taxonomic hierarchy analogous to a biological taxonomy.

FRAMES

AI becomes an industry (1980–present)

The first successful commercial expert system, R1, began operation at the Digital Equipment Corporation (McDermott, 1982). The program helped configure orders for new computer systems; by 1986, it was saving the company an estimated $40 million a year. By 1988, DEC's AI group had 40 expert systems deployed, with more on the way. Du Pont had 100 in use and 500 in development, saving an estimated $10 million a year. Nearly every major U.S. corporation had its own AI group and was either using or investigating expert systems.

In 1981, the Japanese announced the "Fifth Generation" project, a 10-year plan to build intelligent computers running Prolog. In response the United States formed the Microelectronics and Computer Technology Corporation (MCC) as a research consortium designed to assure national competitiveness. In both cases, AI was part of a broad effort, including chip design and human-interface research. However, the AI components of MCC and the Fifth Generation projects never met their ambitious goals. In Britain, the Alvey report reinstated the funding that was cut by the Lighthill report.[15]

Overall, the AI industry boomed from a few million dollars in 1980 to billions of dollars in 1988. Soon after that came a period called the "AI Winter," in which many companies suffered as they failed to deliver on extravagant promises.

[15] To save embarrassment, a new field called IKBS (Intelligent Knowledge-Based Systems) was invented because Artificial Intelligence had been officially canceled.

The return of neural networks (1986–present)

Although computer science had largely abandoned the field of neural networks in the late 1970s, work continued in other fields. Physicists such as John Hopfield (1982) used techniques from statistical mechanics to analyze the storage and optimization properties of networks, treating collections of nodes like collections of atoms. Psychologists including David Rumelhart and Geoff Hinton continued the study of neural-net models of memory. As we discuss in Chapter 20, the real impetus came in the mid-1980s when at least four different groups reinvented the back-propagation learning algorithm first found in 1969 by Bryson and Ho. The algorithm was applied to many learning problems in computer science and psychology, and the widespread dissemination of the results in the collection *Parallel Distributed Processing* (Rumelhart and McClelland, 1986) caused great excitement.

CONNECTIONIST

These so-called **connectionist** models of intelligent systems were seen by some as direct competitors both to the symbolic models promoted by Newell and Simon and to the logicist approach of McCarthy and others (Smolensky, 1988). It might seem obvious that at some level humans manipulate symbols—in fact, Terrence Deacon's book *The Symbolic Species* (1997) suggests that this is the *defining characteristic* of humans, but the most ardent connectionists questioned whether symbol manipulation had any real explanatory role in detailed models of cognition. This question remains unanswered, but the current view is that connectionist and symbolic approaches are complementary, not competing.

AI becomes a science (1987–present)

Recent years have seen a revolution in both the content and the methodology of work in artificial intelligence.[16] It is now more common to build on existing theories than to propose brand new ones, to base claims on rigorous theorems or hard experimental evidence rather than on intuition, and to show relevance to real-world applications rather than toy examples.

AI was founded in part as a rebellion against the limitations of existing fields like control theory and statistics, but now it is embracing those fields. As David McAllester (1998) put it,

> In the early period of AI it seemed plausible that new forms of symbolic computation, e.g., frames and semantic networks, made much of classical theory obsolete. This led to a form of isolationism in which AI became largely separated from the rest of computer science. This isolationism is currently being abandoned. There is a recognition that machine learning should not be isolated from information theory, that uncertain reasoning should not be isolated from stochastic modeling, that search should not be isolated from classical optimization and control, and that automated reasoning should not be isolated from formal methods and static analysis.

In terms of methodology, AI has finally come firmly under the scientific method. To be accepted, hypotheses must be subjected to rigorous empirical experiments, and the results must

[16] Some have characterized this change as a victory of the **neats**—those who think that AI theories should be grounded in mathematical rigor—over the **scruffies**—those who would rather try out lots of ideas, write some programs, and then assess what seems to be working. Both approaches are important. A shift toward neatness implies that the field has reached a level of stability and maturity. Whether that stability will be disrupted by a new scruffy idea is another question.

be analyzed statistically for their importance (Cohen, 1995). Through the use of the Internet and shared repositories of test data and code, it is now possible to replicate experiments.

The field of speech recognition illustrates the pattern. In the 1970s, a wide variety of different architectures and approaches were tried. Many of these were rather *ad hoc* and fragile, and were demonstrated on only a few specially selected examples. In recent years, approaches based on **hidden Markov models** (HMMs) have come to dominate the area. Two aspects of HMMs are relevant. First, they are based on a rigorous mathematical theory. This has allowed speech researchers to build on several decades of mathematical results developed in other fields. Second, they are generated by a process of training on a large corpus of real speech data. This ensures that the performance is robust, and in rigorous blind tests the HMMs have been improving their scores steadily. Speech technology and the related field of handwritten character recognition are already making the transition to widespread industrial and consumer applications.

Neural networks also fit this trend. Much of the work on neural nets in the 1980s was done in an attempt to scope out what could be done and to learn how neural nets differ from "traditional" techniques. Using improved methodology and theoretical frameworks, the field arrived at an understanding in which neural nets can now be compared with corresponding techniques from statistics, pattern recognition, and machine learning, and the most promising technique can be applied to each application. As a result of these developments, so-called **data mining** technology has spawned a vigorous new industry.

DATA MINING

Judea Pearl's (1988) *Probabilistic Reasoning in Intelligent Systems* led to a new acceptance of probability and decision theory in AI, following a resurgence of interest epitomized by Peter Cheeseman's (1985) article "In Defense of Probability." The **Bayesian network** formalism was invented to allow efficient representation of, and rigorous reasoning with, uncertain knowledge. This approach largely overcomes many problems of the probabilistic reasoning systems of the 1960s and 1970s; it now dominates AI research on uncertain reasoning and expert systems. The approach allows for learning from experience, and it combines the best of classical AI and neural nets. Work by Judea Pearl (1982a) and by Eric Horvitz and David Heckerman (Horvitz and Heckerman, 1986; Horvitz *et al.*, 1986) promoted the idea of *normative* expert systems: ones that act rationally according to the laws of decision theory and do not try to imitate the thought steps of human experts. The Windows™ operating system includes several normative diagnostic expert systems for correcting problems. Chapters 13 to 16 cover this area.

Similar gentle revolutions have occurred in robotics, computer vision, and knowledge representation. A better understanding of the problems and their complexity properties, combined with increased mathematical sophistication, has led to workable research agendas and robust methods. In many cases, formalization and specialization have also led to fragmentation: topics such as vision and robotics are increasingly isolated form "mainstream" AI work. The unifying view of AI as rational agent design is one that can bring unity back to these disparate fields.

The emergence of intelligent agents (1995–present)

Perhaps encouraged by the progress in solving the subproblems of AI, researchers have also started to look at the "whole agent" problem again. The work of Allen Newell, John Laird, and Paul Rosenbloom on SOAR (Newell, 1990; Laird *et al.*, 1987) is the best-known example of a complete agent architecture. The so-called situated movement aims to understand the workings of agents embedded in real environments with continuous sensory inputs. One of the most important environments for intelligent agents is the Internet. AI systems have become so common in web-based applications that the "-bot" suffix has entered everyday language. Moreover, AI technologies underlie many Internet tools, such as search engines, recommender systems, and Web site construction systems.

Besides the first edition of this text (Russell and Norvig, 1995), other recent texts have also adopted the agent perspective (Poole *et al.*, 1998; Nilsson, 1998). One consequence of trying to build complete agents is the realization that the previously isolated subfields of AI might need to be reorganized somewhat when their results are to be tied together. In particular, it is now widely appreciated that sensory systems (vision, sonar, speech recognition, etc.) cannot deliver perfectly reliable information about the environment. Hence, reasoning and planning systems must be able to handle uncertainty. A second major consequence of the agent perspective is that AI has been drawn into much closer contact with other fields, such as control theory and economics, that also deal with agents.

1.4 THE STATE OF THE ART

What can AI do today? A concise answer is difficult, because there are so many activities in so many subfields. Here we sample a few applications; others appear throughout the book.

Autonomous planning and scheduling: A hundred million miles from Earth, NASA's Remote Agent program became the first on-board autonomous planning program to control the scheduling of operations for a spacecraft (Jonsson *et al.*, 2000). Remote Agent generated plans from high-level goals specified from the ground, and it monitored the operation of the spacecraft as the plans were executed—detecting, diagnosing, and recovering from problems as they occurred.

Game playing: IBM's Deep Blue became the first computer program to defeat the world champion in a chess match when it bested Garry Kasparov by a score of 3.5 to 2.5 in an exhibition match (Goodman and Keene, 1997). Kasparov said that he felt a "new kind of intelligence" across the board from him. *Newsweek* magazine described the match as "The brain's last stand." The value of IBM's stock increased by $18 billion.

Autonomous control: The ALVINN computer vision system was trained to steer a car to keep it following a lane. It was placed in CMU's NAVLAB computer-controlled minivan and used to navigate across the United States—for 2850 miles it was in control of steering the vehicle 98% of the time. A human took over the other 2%, mostly at exit ramps. NAVLAB has video cameras that transmit road images to ALVINN, which then computes the best direction to steer, based on experience from previous training runs.

Diagnosis: Medical diagnosis programs based on probabilistic analysis have been able to perform at the level of an expert physician in several areas of medicine. Heckerman (1991) describes a case where a leading expert on lymph-node pathology scoffs at a program's diagnosis of an especially difficult case. The creators of the program suggest he ask the computer for an explanation of the diagnosis. The machine points out the major factors influencing its decision and explains the subtle interaction of several of the symptoms in this case. Eventually, the expert agrees with the program.

Logistics Planning: During the Persian Gulf crisis of 1991, U.S. forces deployed a Dynamic Analysis and Replanning Tool, DART (Cross and Walker, 1994), to do automated logistics planning and scheduling for transportation. This involved up to 50,000 vehicles, cargo, and people at a time, and had to account for starting points, destinations, routes, and conflict resolution among all parameters. The AI planning techniques allowed a plan to be generated in hours that would have taken weeks with older methods. The Defense Advanced Research Project Agency (DARPA) stated that this single application more than paid back DARPA's 30-year investment in AI.

Robotics: Many surgeons now use robot assistants in microsurgery. HipNav (DiGioia *et al.*, 1996) is a system that uses computer vision techniques to create a three-dimensional model of a patient's internal anatomy and then uses robotic control to guide the insertion of a hip replacement prosthesis.

Language understanding and problem solving: PROVERB (Littman *et al.*, 1999) is a computer program that solves crossword puzzles better than most humans, using constraints on possible word fillers, a large database of past puzzles, and a variety of information sources including dictionaries and online databases such as a list of movies and the actors that appear in them. For example, it determines that the clue "Nice Story" can be solved by "ETAGE" because its database includes the clue/solution pair "Story in France/ETAGE" and because it recognizes that the patterns "Nice *X*" and "*X* in France" often have the same solution. The program does not know that Nice is a city in France, but it can solve the puzzle.

These are just a few examples of artificial intelligence systems that exist today. Not magic or science fiction—but rather science, engineering, and mathematics, to which this book provides an introduction.

1.5 SUMMARY

This chapter defines AI and establishes the cultural background against which it has developed. Some of the important points are as follows:

- Different people think of AI differently. Two important questions to ask are: Are you concerned with thinking or behavior? Do you want to model humans or work from an ideal standard?
- In this book, we adopt the view that intelligence is concerned mainly with **rational action**. Ideally, an **intelligent agent** takes the best possible action in a situation. We will study the problem of building agents that are intelligent in this sense.

- Philosophers (going back to 400 B.C.) made AI conceivable by considering the ideas that the mind is in some ways like a machine, that it operates on knowledge encoded in some internal language, and that thought can be used to choose what actions to take.
- Mathematicians provided the tools to manipulate statements of logical certainty as well as uncertain, probabilistic statements. They also set the groundwork for understanding computation and reasoning about algorithms.
- Economists formalized the problem of making decisions that maximize the expected outcome to the decision-maker.
- Psychologists adopted the idea that humans and animals can be considered information-processing machines. Linguists showed that language use fits into this model.
- Computer engineers provided the artifacts that make AI applications possible. AI programs tend to be large, and they could not work without the great advances in speed and memory that the computer industry has provided.
- Control theory deals with designing devices that act optimally on the basis of feedback from the environment. Initially, the mathematical tools of control theory were quite different from AI, but the fields are coming closer together.
- The history of AI has had cycles of success, misplaced optimism, and resulting cutbacks in enthusiasm and funding. There have also been cycles of introducing new creative approaches and systematically refining the best ones.
- AI has advanced more rapidly in the past decade because of greater use of the scientific method in experimenting with and comparing approaches.
- Recent progress in understanding the theoretical basis for intelligence has gone hand in hand with improvements in the capabilities of real systems. The subfields of AI have become more integrated, and AI has found common ground with other disciplines.

BIBLIOGRAPHICAL AND HISTORICAL NOTES

The methodological status of artificial intelligence is investigated in *The Sciences of the Artificial*, by Herb Simon (1981), which discusses research areas concerned with complex artifacts. It explains how AI can be viewed as both science and mathematics. Cohen (1995) gives an overview of experimental methodology within AI. Ford and Hayes (1995) give an opinionated view of the usefulness of the Turing Test.

Artificial Intelligence: The Very Idea, by John Haugeland (1985) gives a readable account of the philosophical and practical problems of AI. Cognitive science is well described by several recent texts (Johnson-Laird, 1988; Stillings *et al.*, 1995; Thagard, 1996) and by the *Encyclopedia of the Cognitive Sciences* (Wilson and Keil, 1999). Baker (1989) covers the syntactic part of modern linguistics, and Chierchia and McConnell-Ginet (1990) cover semantics. Jurafsky and Martin (2000) cover computational linguistics.

Early AI is described in Feigenbaum and Feldman's *Computers and Thought* (1963), Minsky's *Semantic Information Processing* (1968), and the *Machine Intelligence* series edited by Donald Michie. A large number of influential papers have been anthologized by Webber

and Nilsson (1981) and by Luger (1995). Early papers on neural networks are collected in *Neurocomputing* (Anderson and Rosenfeld, 1988). The *Encyclopedia of AI* (Shapiro, 1992) contains survey articles on almost every topic in AI. These articles usually provide a good entry point into the research literature on each topic.

The most recent work appears in the proceedings of the major AI conferences: the biennial International Joint Conference on AI (IJCAI), the annual European Conference on AI (ECAI), and the National Conference on AI, more often known as AAAI, after its sponsoring organization. The major journals for general AI are *Artificial Intelligence*, *Computational Intelligence*, the *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *IEEE Intelligent Systems*, and the electronic *Journal of Artificial Intelligence Research*. There are also many conferences and journals devoted to specific areas, which we cover in the appropriate chapters. The main professional societies for AI are the American Association for Artificial Intelligence (AAAI), the ACM Special Interest Group in Artificial Intelligence (SIGART), and the Society for Artificial Intelligence and Simulation of Behaviour (AISB). AAAI's *AI Magazine* contains many topical and tutorial articles, and its website, aaai.org, contains news and background information.

EXERCISES

These exercises are intended to stimulate discussion, and some might be set as term projects. Alternatively, preliminary attempts can be made now, and these attempts can be reviewed after the completion of the book.

1.1 Define in your own words: (a) intelligence, (b) artificial intelligence, (c) agent.

1.2 Read Turing's original paper on AI (Turing, 1950). In the paper, he discusses several potential objections to his proposed enterprise and his test for intelligence. Which objections still carry some weight? Are his refutations valid? Can you think of new objections arising from developments since he wrote the paper? In the paper, he predicts that, by the year 2000, a computer will have a 30% chance of passing a five-minute Turing Test with an unskilled interrogator. What chance do you think a computer would have today? In another 50 years?

1.3 Every year the Loebner prize is awarded to the program that comes closest to passing a version of the Turing test. Research and report on the latest winner of the Loebner prize. What techniques does it use? How does it advance the state of the art in AI?

1.4 There are well-known classes of problems that are intractably difficult for computers, and other classes that are provably undecidable. Does this mean that AI is impossible?

1.5 Suppose we extend Evans's ANALOGY program so that it can score 200 on a standard IQ test. Would we then have a program more intelligent than a human? Explain.

1.6 How could introspection—reporting on one's inner thoughts—be inaccurate? Could I be wrong about what I'm thinking? Discuss.

1.7 Examine the AI literature to discover whether the following tasks can currently be solved by computers:

a. Playing a decent game of table tennis (ping-pong).
b. Driving in the center of Cairo.
c. Buying a week's worth of groceries at the market.
d. Buying a week's worth of groceries on the web.
e. Playing a decent game of bridge at a competitive level.
f. Discovering and proving new mathematical theorems.
g. Writing an intentionally funny story.
h. Giving competent legal advice in a specialized area of law.
i. Translating spoken English into spoken Swedish in real time.
j. Performing a complex surgical operation.

For the currently infeasible tasks, try to find out what the difficulties are and predict when, if ever, they will be overcome.

1.8 Some authors have claimed that perception and motor skills are the most important part of intelligence, and that "higher level" capacities are necessarily parasitic—simple add-ons to these underlying facilities. Certainly, most of evolution and a large part of the brain have been devoted to perception and motor skills, whereas AI has found tasks such as game playing and logical inference to be easier, in many ways, than perceiving and acting in the real world. Do you think that AI's traditional focus on higher-level cognitive abilities is misplaced?

1.9 Why would evolution tend to result in systems that act rationally? What goals are such systems designed to achieve?

1.10 Are reflex actions (such as moving your hand away from a hot stove) rational? Are they intelligent?

1.11 "Surely computers cannot be intelligent—they can do only what their programmers tell them." Is the latter statement true, and does it imply the former?

1.12 "Surely animals cannot be intelligent—they can do only what their genes tell them." Is the latter statement true, and does it imply the former?

1.13 "Surely animals, humans, and computers cannot be intelligent—they can do only what their constituent atoms are told to do by the laws of physics." Is the latter statement true, and does it imply the former?

CHAPTER 27

PHILOSOPHY, ETHICS, AND SAFETY OF AI

In which we consider the big questions around the meaning of AI, how we can ethically develop and apply it, and how we can keep it safe.

Philosophers have been asking big questions for a long time: How do minds work? Is it possible for machines to act intelligently in the way that people do? Would such machines have real, conscious minds?

To these, we add new ones: What are the ethical implications of intelligent machines in day-to-day use? Should machines be allowed to decide to kill humans? Can algorithms be fair and unbiased? What will humans do if machines can do all kinds of work? And how do we control machines that may become more intelligent than us?

27.1 The Limits of AI

In 1980, philosopher John Searle introduced a distinction between **weak AI**—the idea that machines could act *as if* they were intelligent—and **strong AI**—the assertion that machines that do so are *actually* consciously thinking (not just *simulating* thinking). Over time the definition of strong AI shifted to refer to what is also called "human-level AI" or "general AI"—programs that can solve an arbitrarily wide variety of tasks, including novel ones, and do so as well as a human.

Weak AI
Strong AI

Critics of weak AI who objected to the very possibility of intelligent behavior in machines now appear as shortsighted as Simon Newcomb, who in October 1903 wrote "aerial flight is one of the great class of problems with which man can never cope"—just two months before the Wright brothers' flight at Kitty Hawk. The rapid progress of recent years does not, however, prove that there can be no limits to what AI can achieve. Alan Turing (1950), the first person to define AI, was also the first to raise possible objections to AI, foreseeing almost all the ones subsequently raised by others.

27.1.1 The argument from informality

Turing's "argument from informality of behavior" says that human behavior is far too complex to be captured by any formal set of rules—humans must be using some informal guidelines that (the argument claims) could never be captured in a formal set of rules and thus could never be codified in a computer program.

A key proponent of this view was Hubert Dreyfus, who produced a series of influential critiques of artificial intelligence: *What Computers Can't Do* (1972), the sequel *What*

Computers Still Can't Do (1992), and, with his brother Stuart, *Mind Over Machine* (1986). Similarly, philosopher Kenneth Sayre (1993) said "Artificial intelligence *pursued within the cult of computationalism* stands not even a ghost of a chance of producing durable results." The technology they criticize came to be called **Good Old-Fashioned AI (GOFAI)**.

Good Old-Fashioned AI (GOFAI)

GOFAI corresponds to the simplest logical agent design described in Chapter 7, and we saw there that it is indeed difficult to capture every contingency of appropriate behavior in a set of necessary and sufficient logical rules; we called that the **qualification problem**. But as we saw in Chapter 12, probabilistic reasoning systems are more appropriate for open-ended domains, and as we saw in Chapter 21, deep learning systems do well on a variety of "informal" tasks. Thus, the critique is not addressed against computers *per se*, but rather against one particular style of programming them with logical rules—a style that was popular in the 1980s but has been eclipsed by new approaches.

One of Dreyfus's strongest arguments is for situated agents rather than disembodied logical inference engines. An agent whose understanding of "dog" comes only from a limited set of logical sentences such as "$Dog(x) \Rightarrow Mammal(x)$" is at a disadvantage compared to an agent that has watched dogs run, has played fetch with them, and has been licked by one. As philosopher Andy Clark (1998) says, "Biological brains are first and foremost the control systems for biological bodies. Biological bodies move and act in rich real-world surroundings." According to Clark, we are "good at frisbee, bad at logic."

Embodied cognition

The **embodied cognition** approach claims that it makes no sense to consider the brain separately: cognition takes place within a body, which is embedded in an environment. We need to study the system as a whole; the brain's functioning exploits regularities in its environment, including the rest of its body. Under the embodied cognition approach, robotics, vision, and other sensors become central, not peripheral.

Overall, Dreyfus saw areas where AI did not have complete answers and said that AI is therefore impossible; we now see many of these same areas undergoing continued research and development leading to increased capability, not impossibility.

27.1.2 The argument from disability

The "argument from disability" makes the claim that "a machine can never do *X*." As examples of *X*, Turing lists the following:

> Be kind, resourceful, beautiful, friendly, have initiative, have a sense of humor, tell right from wrong, make mistakes, fall in love, enjoy strawberries and cream, make someone fall in love with it, learn from experience, use words properly, be the subject of its own thought, have as much diversity of behavior as man, do something really new.

In retrospect, some of these are rather easy—we're all familiar with computers that "make mistakes." Computers with metareasoning capabilities (Chapter 5) can examine heir own computations, thus being the subject of their own reasoning. A century-old technology has the proven ability to "make someone fall in love with it"—the teddy bear. Computer chess expert David Levy predicts that by 2050 people will routinely fall in love with humanoid robots. As for a robot falling in love, that is a common theme in fiction,[1] but there has been only limited academic speculation on the subject (Kim *et al.*, 2007). Computers have

[1] For example, the opera Coppélia (1870), the novel *Do Androids Dream of Electric Sheep?* (1968), the movies *AI* (2001), *Wall-E* (2008), and *Her* (2013).

done things that are "really new," making significant discoveries in astronomy, mathematics, chemistry, mineralogy, biology, computer science, and other fields, and creating new forms of art through style transfer (Gatys *et al.*, 2016). Overall, programs exceed human performance in some tasks and lag behind on others. The one thing that it is clear they can't do is be exactly human.

27.1.3 The mathematical objection

Turing (1936) and Gödel (1931) proved that certain mathematical questions are in principle unanswerable by particular formal systems. Gödel's incompleteness theorem (see Section 9.5) is the most famous example of this. Briefly, for any formal axiomatic framework F powerful enough to do arithmetic, it is possible to construct a so-called Gödel sentence $G(F)$ with the following properties:

- $G(F)$ is a sentence of F, but cannot be proved within F.
- If F is consistent, then $G(F)$ is true.

Philosophers such as J. R. Lucas (1961) have claimed that this theorem shows that machines are mentally inferior to humans, because machines are formal systems that are limited by the incompleteness theorem—they cannot establish the truth of their own Gödel sentence—while humans have no such limitation. This has caused a lot of controversy, spawning a vast literature, including two books by the mathematician/physicist Sir Roger Penrose (1989, 1994). Penrose repeats Lucas's claim with some fresh twists, such as the hypothesis that humans are different because their brains operate by quantum gravity—a theory that makes multiple false predictions about brain physiology.

We will examine three of the problems with Lucas's claim. First, an agent should not be ashamed that it cannot establish the truth of some sentence while other agents can. Consider the following sentence:

> Lucas cannot consistently assert that this sentence is true.

If Lucas asserted this sentence, then he would be contradicting himself, so therefore Lucas cannot consistently assert it, and hence it is true. We have thus demonstrated that there is a true sentence that Lucas cannot consistently assert while other people (and machines) can. But that does not make us think any less of Lucas.

Second, Gödel's incompleteness theorem and related results apply to *mathematics*, not to *computers*. No entity—human or machine—can prove things that are impossible to prove. Lucas and Penrose falsely assume that humans can somehow get around these limits, as when Lucas (1976) says "we must assume our own consistency, if thought is to be possible at all." But this is an unwarranted assumption: humans are notoriously inconsistent. This is certainly true for everyday reasoning, but it is also true for careful mathematical thought. A famous example is the four-color map problem. Alfred Kempe (1879) published a proof that was widely accepted for 11 years until Percy Heawood (1890) pointed out a flaw.

Third, Gödel's incompleteness theorem technically applies only to formal systems that are powerful enough to do arithmetic. This includes Turing machines, and Lucas's claim is in part based on the assertion that computers are equivalent to Turing machines. This is not quite true. Turing machines are infinite, whereas computers (and brains) are finite, and any computer can therefore be described as a (very large) system in propositional logic, which is not subject to Gödel's incompleteness theorem. Lucas assumes that humans can "change their

minds" while computers cannot, but that is also false—a computer can retract a conclusion after new evidence or further deliberation; it can upgrade its hardware; and it can change its decision-making processes with machine learning or software rewriting.

27.1.4 Measuring AI

Alan Turing, in his famous paper "Computing Machinery and Intelligence" (1950), suggested that instead of asking whether machines can think, we should ask whether machines can pass a behavioral test, which has come to be called the **Turing test**. The test requires a program to have a conversation (via typed messages) with an interrogator for five minutes. The interrogator then has to guess if the conversation is with a program or a person; the program passes the test if it fools the interrogator 30% of the time. To Turing, the key point was not the exact details of the test, but instead the idea of measuring intelligence by performance on some kind of open-ended behavioral task, rather than by philosophical speculation.

Nevertheless, Turing conjectured that by the year 2000 a computer with a storage of a billion units could pass the test, but here we are on the other side of 2000, and we still can't agree whether any program has passed. Many people have been fooled when they didn't know they might be chatting with a computer. The ELIZA program and Internet chatbots such as MGONZ (Humphrys, 2008) and NATACHATA (Jonathan *et al.*, 2009) fool their correspondents repeatedly, and the chatbot CYBERLOVER has attracted the attention of law enforcement because of its penchant for tricking fellow chatters into divulging enough personal information that their identity can be stolen.

In 2014, a chatbot called Eugene Goostman fooled 33% of the untrained amateur judges in a Turing test. The program claimed to be a boy from Ukraine with limited command of English; this helped explain its grammatical errors. Perhaps the Turing test is really a test of human gullibility. So far no well-trained judge has been fooled (Aaronson, 2014).

Turing test competitions have led to better chatbots, but have not been a focus of research within the AI community. Instead, AI researchers who crave competition are more likely to concentrate on playing chess or Go or StarCraft II, or taking an 8th grade science exam, or identifying objects in images. In many of these competitions, programs have reached or surpassed human-level performance, but that doesn't mean the programs are human-like outside the specific task. The point is to improve basic science and technology and to provide useful tools, not to fool judges.

27.2 Can Machines Really Think?

Some philosophers claim that a machine that acts intelligently would not be *actually* thinking, but would be only a *simulation* of thinking. But most AI researchers are not concerned with the distinction, and the computer scientist Edsger Dijkstra (1984) said that "The question of whether *Machines Can Think* . . . is about as relevant as the question of whether *Submarines Can Swim*." The American Heritage Dictionary's first definition of *swim* is "To move through water by means of the limbs, fins, or tail," and most people agree that submarines, being limbless, cannot swim. The dictionary also defines *fly* as "To move through the air by means of wings or winglike parts," and most people agree that airplanes, having winglike parts, can fly. However, neither the questions nor the answers have any relevance to the design or capabilities of airplanes and submarines; rather they are about word usage in English. (The

fact that ships do *swim* (*"privet"*) in Russian amplifies this point.) English speakers have not yet settled on a precise definition for the word "think"—does it require "a brain" or just "brain-like parts?"

Again, the issue was addressed by Turing. He notes that we never have *any* direct evidence about the internal mental states of other humans—a kind of mental solipsism. Nevertheless, Turing says, "Instead of arguing continually over this point, it is usual to have the **polite convention** that everyone thinks." Turing argues that we would also extend the polite convention to machines, if only we had experience with ones that act intelligently. However, now that we do have some experience, it seems that our willingness to ascribe sentience depends at least as much on humanoid appearance and voice as on pure intelligence.

Polite convention

27.2.1 The Chinese room

The philosopher John Searle rejects the polite convention. His famous **Chinese room** argument (Searle, 1990) goes as follows: Imagine a human, who understands only English, inside a room that contains a rule book, written in English, and various stacks of paper. Pieces of paper containing indecipherable symbols are slipped under the door to the room. The human follows the instructions in the rule book, finding symbols in the stacks, writing symbols on new pieces of paper, rearranging the stacks, and so on. Eventually, the instructions will cause one or more symbols to be transcribed onto a piece of paper that is passed back to the outside world. From the outside, we see a system that is taking input in the form of Chinese sentences and generating fluent, intelligent Chinese responses.

Chinese room

Searle then argues: it is given that the human does not understand Chinese. The rule book and the stacks of paper, being just pieces of paper, do not understand Chinese. Therefore, there is no understanding of Chinese. And Searle says that the Chinese room is doing the same thing that a computer would do, so therefore computers generate no understanding.

Searle (1980) is a proponent of **biological naturalism**, according to which mental states are high-level emergent features that are caused by low-level physical processes *in the neurons*, and it is the (unspecified) properties of the neurons that matter: according to Searle's biases, neurons have "it" and transistors do not. There have been many refutations of Searle's argument, but no consensus. His argument could equally well be used (perhaps by robots) to argue that a human cannot have true understanding; after all, a human is made out of cells, the cells do not understand, therefore there is no understanding. In fact, that is the plot of Terry Bisson's (1990) science fiction story *They're Made Out of Meat*, in which alien robots explore Earth and can't believe that hunks of meat could possibly be sentient. How they can be remains a mystery.

Biological naturalism

27.2.2 Consciousness and qualia

Running through all the debates about strong AI is the issue of **consciousness**: awareness of the outside world, and of the self, and the subjective experience of living. The technical term for the intrinsic nature of experiences is **qualia** (from the Latin word meaning, roughly, "of what kind"). The big question is whether machines can have qualia. In the movie *2001*, when astronaut David Bowman is disconnecting the "cognitive circuits" of the HAL 9000 computer, it says *"I'm afraid, Dave. Dave, my mind is going. I can feel it."* Does HAL actually have feelings (and deserve sympathy)? Or is the reply just an algorithmic response, no different from "Error 404: not found"?

Consciousness

Qualia

There is a similar question for animals: pet owners are certain that their dog or cat has consciousness, but not all scientists agree. Crickets change their behavior based on temperature, but few people would say that crickets experience the *feeling* of being warm or cold.

One reason that the problem of consciousness is hard is that it remains ill-defined, even after centuries of debate. But help may be on the way. Recently philosophers have teamed with neuroscientists under the auspices of the Templeton Foundation to start a series of experiments that could resolve some of the issues. Advocates of two leading theories of consciousness (global workspace theory and integrated information theory) have agreed that the experiments could confirm one theory over the other—a rarity in philosophy.

Alan Turing (1950) concedes that the question of consciousness is a difficult one, but denies that it has much relevance to the practice of AI: "I do not wish to give the impression that I think there is no mystery about consciousness ... But I do not think these mysteries necessarily need to be solved before we can answer the question with which we are concerned in this paper." We agree with Turing—we are interested in creating programs that behave intelligently. Individual aspects of consciousness—awareness, self-awareness, attention—can be programmed and can be part of an intelligent machine. The additional project of making a machine conscious in exactly the way humans are is not one that we are equipped to take on. We do agree that behaving intelligently will require some degree of *awareness*, which will differ from task to task, and that tasks involving interaction with humans will require a model of human subjective experience.

In the matter of modeling experience, humans have a clear advantage over machines, because they can use their own subjective apparatus to appreciate the subjective experience of others. For example, if you want to know *what it's like* when someone hits their thumb with a hammer, you can hit your thumb with a hammer. Machines have no such capability—although unlike humans, they can run each other's code.

27.3 The Ethics of AI

Given that AI is a powerful technology, we have a moral obligation to use it well, to promote the positive aspects and avoid or mitigate the negative ones.

The positive aspects are many. For example, AI can save lives through improved medical diagnosis, new medical discoveries, better prediction of extreme weather events, and safer driving with driver assistance and (eventually) self-driving technologies. There are also many opportunities to improve lives. Microsoft's AI for Humanitarian Action program applies AI to recovering from natural disasters, addressing the needs of children, protecting refugees, and promoting human rights. Google's AI for Social Good program supports work on rainforest protection, human rights jurisprudence, pollution monitoring, measurement of fossil fuel emissions, crisis counseling, news fact checking, suicide prevention, recycling, and other issues. The University of Chicago's Center for Data Science for Social Good applies machine learning to problems in criminal justice, economic development, education, public health, energy, and environment.

AI applications in crop management and food production help feed the world. Optimization of business processes using machine learning will make businesses more productive, increasing wealth and providing more employment. Automation can replace the tedious and dangerous tasks that many workers face, and free them to concentrate on more interesting

aspects. People with disabilities will benefit from AI-based assistance in seeing, hearing, and mobility. Machine translation already allows people from different cultures to communicate. Software-based AI solutions have near zero marginal cost of production, and so have the potential to democratize access to advanced technology (even as other aspects of software have the potential to centralize power).

Despite these many positive aspects, we shouldn't ignore the negatives. Many new technologies have had unintended **negative side effects**: nuclear fission brought Chernobyl and the threat of global destruction; the internal combustion engine brought air pollution, global warming, and the paving of paradise. Other technologies can have negative effects even when used as intended, such as sarin gas, AR-15 rifles, and telephone solicitation. Automation will create wealth, but under current economic conditions much of that wealth will flow to the owners of the automated systems, leading to increased income inequality. This can be disruptive to a well-functioning society. In developing countries, the traditional path to growth through low-cost manufacturing for export may be cut off, as wealthy countries adopt fully automated manufacturing facilities on-shore. Our ethical and governance decisions will dictate the level of inequality that AI will engender.

Negative side effects

All scientists and engineers face ethical considerations of what projects they should or should not take on, and how they can make sure the execution of the project is safe and beneficial. In 2010, the UK's Engineering and Physical Sciences Research Council held a meeting to develop a set of Principles of Robotics. In subsequent years other government agencies, nonprofit organizations, and companies created similar sets of principles. The gist is that every organization that creates AI technology, and everyone in the organization, has a responsibility to make sure the technology contributes to good, not harm. The most commonly-cited principles are:

Ensure safety	Establish accountability
Ensure fairness	Uphold human rights and values
Respect privacy	Reflect diversity/inclusion
Promote collaboration	Avoid concentration of power
Provide transparency	Acknowledge legal/policy implications
Limit harmful uses of AI	Contemplate implications for employment

Note that many of the principles, such as "ensure safety," have applicability to all software or hardware systems, not just AI systems. Several principles are worded in a vague way, making them difficult to measure or enforce. That is in part because AI is a big field with many subfields, each of which has a different set of historical norms and different relationships between the AI developers and the stakeholders. Mittelstadt (2019) suggests that the subfields should each develop more specific actionable guidelines and case precedents.

27.3.1 Lethal autonomous weapons

The UN defines a lethal autonomous weapon as one that locates, selects, and engages (i.e., kills) human targets without human supervision. Various weapons fulfill some of these criteria. For example, land mines have been used since the 17th century: they can select and engage targets in a limited sense according to the degree of pressure exerted or the quantity of metal present, but they cannot go out and locate targets by themselves. (Land mines are banned under the Ottawa Treaty.) Guided missiles, in use since the 1940s, can chase targets, but they have to be pointed in the right general direction by a human. Auto-firing

radar-controlled guns have been used to defend naval ships since the 1970s; they are mainly intended to destroy incoming missiles, but they could also attack manned aircraft. Although the word "autonomous" is often used to describe unmanned air vehicles or **drones**, most such weapons are both remotely piloted and require human actuation of the lethal payload.

At the time of writing, several weapons systems seem to have crossed the line into full autonomy. For example Israel's Harop missile is a "loitering munition" with a ten-foot wingspan and a fifty-pound warhead. It searches for up to six hours in a given geographical region for any target that meets a given criterion and then destroys it. The criterion could be "emits a radar signal resembling antiaircraft radar" or "looks like a tank." The Turkish manufacturer STM advertises its Kargu quadcopter—which carries up to 1.5kg of explosives—as capable of "Autonomous hit . . . targets selected on images . . . tracking moving targets . . . anti-personnel . . . face recognition."

Autonomous weapons have been called the "third revolution in warfare" after gunpowder and nuclear weapons. Their military potential is obvious. For example, few experts doubt that autonomous fighter aircraft would defeat any human pilot. Autonomous aircraft, tanks, and submarines can be cheaper, faster, more maneuverable, and have longer range than their manned counterparts.

Since 2014, the United Nations in Geneva has conducted regular discussions under the auspices of the Convention on Certain Conventional Weapons (CCW) on the question of whether to ban lethal autonomous weapons. At the time of writing, 30 nations, ranging in size from China to the Holy See, have declared their support for an international treaty, while other key countries—including Israel, Russia, South Korea, and the United States—are opposed to a ban.

The debate over autonomous weapons includes legal, ethical and practical aspects. The legal issues are governed primarily by the CCW, which requires the possibility of discriminating between combatants and non-combatants, the judgment of military necessity for an attack, and the assessment of proportionality between the military value of a target and the possibility of collateral damage. The feasibility of meeting these criteria is an engineering question—one whose answer will undoubtedly change over time. At present, discrimination seems feasible in some circumstances and will undoubtedly improve rapidly, but necessity and proportionality are not presently feasible: they require that machines make subjective and situational judgments that are considerably more difficult than the relatively simple tasks of searching for and engaging potential targets. For these reasons, it would be legal to use autonomous weapons only in circumstances where a human operator can reasonably predict that the execution of the mission will not result in civilians being targeted or the weapons conducting unnecessary or disproportionate attacks. This means that, for the time being, only very restricted missions could be undertaken by autonomous weapons.

On the ethical side, some find it simply morally unacceptable to delegate the decision to kill humans to a machine. For example, Germany's ambassador in Geneva has stated that it "will not accept that the decision over life and death is taken solely by an autonomous system" while Japan "has no plan to develop robots with humans out of the loop, which may be capable of committing murder." Gen. Paul Selva, at the time the second-ranking military officer in the United States, said in 2017, "I don't think it's reasonable for us to put robots in charge of whether or not we take a human life." Finally, António Guterres, the head of the United Nations, stated in 2019 that "machines with the power and discretion to take lives

without human involvement are politically unacceptable, morally repugnant and should be prohibited by international law."

More than 140 NGOs in over 60 countries are part of the Campaign to Stop Killer Robots, and an open letter organized in 2015 by the Future of Life Institute organized an open letter was signed by over 4,000 AI researchers[2] and 22,000 others.

Against this, it can be argued that as technology improves it ought to be possible to develop weapons that are *less* likely than human soldiers or pilots to cause civilian casualties. (There is also the important benefit that autonomous weapons reduce the need for human soldiers and pilots to risk death.) Autonomous systems will not succumb to fatigue, frustration, hysteria, fear, anger, or revenge, and need not "shoot first, ask questions later" (Arkin, 2015). Just as guided munitions have reduced collateral damage compared to unguided bombs, one may expect intelligent weapons to further improve the precision of attacks. (Against this, see Benjamin (2013) for an analysis of drone warfare casualties.) This, apparently, is the position of the United States in the latest round of negotiations in Geneva.

Perhaps counterintuitively, the United States is also one of the few nations whose own policies currently preclude the use of autonomous weapons. The 2011 U.S. Department of Defense (DOD) roadmap says: "For the foreseeable future, decisions over the use of force [by autonomous systems] and the choice of which individual targets to engage with lethal force will be retained under human control." The primary reason for this policy is practical: autonomous systems are not reliable enough to be trusted with military decisions.

The issue of reliability came to the fore on September 26, 1983, when Soviet missile officer Stanislav Petrov's computer display flashed an alert of an incoming missile attack. According to protocol, Petrov should have initiated a nuclear counterattack, but he suspected the alert was a bug and treated it as such. He was correct, and World War III was (narrowly) averted. We don't know what would have happened if there had been no human in the loop.

Reliability is a very serious concern for military commanders, who know well the complexity of battlefield situations. Machine learning systems that operate flawlessly in training may perform poorly when deployed. Cyberattacks against autonomous weapons could result in friendly-fire casualties; disconnecting the weapon from all communication may prevent that (assuming it has not already been compromised), but then the weapon cannot be recalled if it is malfunctioning.

The overriding practical issue with autonomous weapons is that they they are scalable weapons of mass destruction, in the sense that the scale of an attack that can be launched is proportional to the amount of hardware one can afford to deploy. A quadcopter two inches in diameter can carry a lethal explosive charge, and one million can fit in a regular shipping container. Precisely because they are autonomous, these weapons would not need one million human supervisors to do their work.

As weapons of mass destruction, scalable autonomous weapons have advantages for the attacker compared to nuclear weapons and carpet bombing: they leave property intact and can be applied selectively to eliminate only those who might threaten an occupying force. They could certainly be used to wipe out an entire ethnic group or all the adherents of a particular religion. In many situations, they would also be untraceable. These characteristics make them particularly attractive to non-state actors.

[2] Including the two authors of this book.

These considerations—particularly those characteristics that advantage the attacker—suggest that autonomous weapons will reduce global and national security for all parties. The rational response for governments seems to be to engage in arms control discussions rather than an arms race.

Dual use

The process of designing a treaty is not without its difficulties, however. AI is a **dual use** technology: AI technologies that have peaceful applications such as flight control, visual tracking, mapping, navigation, and multiagent planning, can easily be applied to military purposes. It is easy to turn an autonomous quadcopter into a weapon simply by attaching an explosive and commanding it to seek out a target. Dealing with this will require careful implementation of compliance regimes with industry cooperation, as has already been demonstrated with some success by the Chemical Weapons Convention.

27.3.2 Surveillance, security, and privacy

In 1976, Joseph Weizenbaum warned that automated speech recognition technology could lead to widespread wiretapping, and hence to a loss of civil liberties. Today, that threat has been realized, with most electronic communication going through central servers that can be monitored, and cities packed with microphones and cameras that can identify and track individuals based on their voice, face, and gait. Surveillance that used to require expensive and scarce human resources can now be done at a mass scale by machines.

Surveillance camera

As of 2018, there were as many as 350 million **surveillance cameras** in China and 70 million in the United States. China and other countries have begun exporting surveillance technology to low-tech countries, some with reputations for mistreating their citizens and disproportionately targeting marginalized communities. AI engineers should be clear on what uses of surveillance are compatible with human rights, and decline to work on applications that are incompatible.

Cybersecurity

As more of our institutions operate online, we become more vulnerable to cybercrime (phishing, credit card fraud, botnets, ransomware) and cyberterrorism (including potentially deadly attacks such as shutting down hospitals and power plants or commandeering self-driving cars). Machine learning can be a powerful tool for both sides in the **cybersecurity** battle. Attackers can use automation to probe for insecurities and they can apply reinforcement learning for phishing attempts and automated blackmail. Defenders can use unsupervised learning to detect anomalous incoming traffic patterns (Chandola *et al.*, 2009; Malhotra *et al.*, 2015) and various machine learning techniques to detect fraud (Fawcett and Provost, 1997; Bolton and Hand, 2002). As attacks get more sophisticated, there is a greater responsibility for all engineers, not just the security experts, to design secure systems from the start. One forecast (Kanal, 2017) puts the market for machine learning in cybersecurity at about $100 billion by 2021.

As we interact with computers for increasing amounts of our daily lives, more data on us is being collected by governments and corporations. Data collectors have a moral and legal responsibility to be good stewards of the data they hold. In the U.S., the Health Insurance Portability and Accountability Act (HIPAA) and the Family Educational Rights and Privacy Act (FERPA) protect the privacy of medical and student records. The European Union's General Data Protection Regulation (GDPR) mandates that companies design their systems with protection of data in mind and requires that they obtain user consent for any collection or processing of data.

Balanced against the individual's right to privacy is the value that society gains from sharing data. We want to be able to stop terrorists without oppressing peaceful dissent, and we want to cure diseases without compromising any individual's right to keep their health history private. One key practice is **de-identification**: eliminating personally identifying information (such as name and social security number) so that medical researchers can use the data to advance the common good. The problem is that the shared de-identified data may be subject to re-identification. For example, if the data strips out the name, social security number, and street address, but includes date of birth, gender, and zip code, then, as shown by Latanya Sweeney (2000), 87% of the U.S. population can be uniquely re-identified. Sweeney emphasized this point by re-identifying the health record for the governor of her state when he was admitted to the hospital. In the **Netflix Prize** competition, de-identified records of individual movie ratings were released, and competitors were asked to come up with a machine learning algorithm that could accurately predict which movies an individual would like. But researchers were able to re-identify individual users by matching the date of a rating in the Netflix database with the date of a similar ranking in the Internet Movie Database (IMDB), where users sometimes use their actual names (Narayanan and Shmatikov, 2006).

De-identification

Netflix Prize

This risk can be mitigated somewhat by **generalizing fields**: for example, replacing the exact birth date with just the year of birth, or a broader range like "20-30 years old." Deleting a field altogether can be seen as a form of generalizing to "any." But generalization alone does not guarantee that records are safe from re-identification; it may be that there is only one person in zip code 94720 who is 90–100 years old. A useful property is **k-anonymity**: a database is k-anonymized if every record in the database is indistinguishable from at least $k-1$ other records. If there are records that are more unique than this, they would have to be further generalized.

K-anonymity

An alternative to sharing de-identified records is to keep all records private, but allow **aggregate querying**. An API for queries against the database is provided, and valid queries receive a response that summarizes the data with a count or average. But no response is given if it would violate certain guarantees of privacy. For example, we could allow an epidemiologist to ask, for each zip code, the percentage of people with cancer. For zip codes with at least n people a percentage would be given (with a small amount of random noise), but no response would be given for zip codes with fewer than n people..

Aggregate querying

Care must be taken to protect against de-identification using multiple queries. For example, if the query "average salary and number of employees of XYZ company age 30-40" gives the response [$81,234, 12] and the query "average salary and number of employees of XYZ company age 30-41" gives the response [$81,199, 13], and if we use LinkedIn to find the one 41-year-old at XYZ company, then we have successfully identified them, and can compute their exact salary, even though all the responses involved 12 or more people. The system must be carefully designed to protect against this, with a combination of limits on the queries that can be asked (perhaps only a predefined set of non-overlapping age ranges can be queried) and the precision of the results (perhaps both queries give the answer "about $81,000").

A stronger guarantee is **differential privacy**, which assures that an attacker cannot use queries to re-identify any individual in the database, even if the attacker can make multiple queries and has access to separate linking databases. The query response employs a randomized algorithm that adds a small amount of noise to the result. Given a database D, any record in the database r, any query Q, and a possible response y to the query, we say that the database

Differential privacy

D has ϵ–differential privacy if the log probability of the response y varies by less than ϵ when we add the record r:

$$|\log P(Q(D)=y) - \log P(Q(D+r)=y)| \leq \epsilon .$$

In other words, whether any one person decides to participate in the data base or not makes no appreciable difference to the answers anyone can get, and therefore there is no privacy disincentive to participate. Many databases are designed to guarantee differential privacy.

Federated learning

So far we have considered the issue of sharing de-identified data from a central database. An approach called **federated learning** (Konečnỳ *et al.*, 2016) has no central database; instead, users maintain their own local databases that keep their data private. However, they can share parameters of a machine learning model that is enhanced with their data, without the risk of revealing any of the private data. Imagine a speech understanding application that users can run locally on their phone. The application contains a baseline neural network, which is then improved by local training on the words that are heard on the user's phone. Periodically, the owners of the application poll a subset of the users and ask them for the parameter values of their improved local network, but not for any of their raw data. The parameter values are combined together to form a new improved model which is then made available to all users, so that they all get the benefit of the training that is done by other users.

Secure aggregation

For this scheme to preserve privacy, we have to be able to guarantee that the model parameters shared by each user cannot be reverse-engineered. If we sent the raw parameters, there is a chance that an adversary inspecting them could deduce whether, say, a certain word had been heard by the user's phone. One way to eliminate this risk is with **secure aggregation** (Bonawitz *et al.*, 2017). The idea is that the central server doesn't need to know the exact parameter value from each distributed user; it only needs to know the average value for each parameter, over all polled users. So each user can disguise their parameter values by adding a unique mask to each value; as long as the sum of the masks is zero, the central server will be able to compute the correct average. Details of the protocol make sure that it is efficient in terms of communication (less than half the bits transmitted correspond to masking), is robust to individual users failing to respond, and is secure in the face of adversarial users, eavesdroppers, or even an adversarial central server.

27.3.3 Fairness and bias

Societal bias

Machine learning is augmenting and sometimes replacing human decision-making in important situations: whose loan gets approved, to what neighborhoods police officers are deployed, who gets pretrial release or parole. But machine learning models can perpetuate **societal bias**. Consider the example of an algorithm to predict whether criminal defendants are likely to re-offend, and thus whether they should be released before trial. It could well be that such a system picks up the racial or gender prejudices of human judges from the examples in the training set. Designers of machine learning systems have a moral responsibility to ensure that their systems are in fact fair. In regulated domains such as credit, education, employment, and housing, they have a legal responsibility as well. But what is fairness? There are multiple criteria; here are six of the most commonly-used concepts:

- **Individual fairness**: A requirement that individuals are treated similarly to other similar individuals, regardless of what class they are in.

- **Group fairness**: A requirement that two classes are treated similarly, as measured by some summary statistic.
- **Fairness through unawareness**: If we delete the race and gender attributes from the data set, then it might seem that the system cannot discriminate on those attributes. Unfortunately, we know that machine learning models can predict latent variables (such as race and gender) given other correlated variables (such as zip code and occupation). Furthermore, deleting those attributes makes it impossible to verify equal opportunity or equal outcomes. Still, some countries (e.g., Germany) have chosen this approach for their demographic statistics (whether or not machine learning models are involved).
- **Equal outcome**: The idea that each demographic class gets the same results; they have **demographic parity**. For example, suppose we have to decide whether we should approve loan applications; the goal is to approve those applicants who will pay back the loan and not those who will default on the loan. Demographic parity says that both males and females should have the same percentage of loans approved. Note that this is a group fairness criterion that does nothing to ensure individual fairness; a well-qualified applicant might be denied and a poorly-qualified applicant might be approved, as long as the overall percentages are equal. Also, this approach favors redress of past biases over accuracy of prediction. If a man and a woman are equal in every way, except the woman receives a lower salary for the same job, should she be approved because she would be equal if not for historical biases, or should she be denied because the lower salary does in fact make her more likely to default?

Demographic parity

- **Equal opportunity**: The idea that the people who truly have the ability to pay back the loan should have an equal chance of being correctly classified as such, regardless of their sex. This approach is also called "balance." It can lead to unequal outcomes and ignores the effect of bias in the societal processes that produced the training data.
- **Equal impact**: People with similar likelihood to pay back the loan should have the same expected utility, regardless of the class they belong to. This goes beyond equal opportunity in that it considers both the benefits of a true prediction and the costs of a false prediction.

Let us examine how these issues play out in a particular context. COMPAS is a commercial system for recidivism (re-offense) scoring. It assigns to a defendant in a criminal case a **risk score**, which is then used by a judge to help make decisions: Is it safe to release the defendant before trial, or should they be held in jail? If convicted, how long should the sentence be? Should parole be granted? Given the significance of these decisions, the system has been the subject of intense scrutiny (Dressel and Farid, 2018).

COMPAS is designed to be **well calibrated**: all the individuals who are given the same score by the algorithm should have approximately the same probability of re-offending, regardless of race. For example, among all people that the model assigns a risk score of 7 out of 10, 60% of whites and 61% of blacks re-offend. The designers thus claim that it meets the desired fairness goal.

Well calibrated

On the other hand, COMPAS does not achieve equal opportunity: the proportion of those who did not re-offend but were falsely rated as high-risk was 45% for blacks and 23% for whites. In the case *State v. Loomis*, where a judge relied on COMPAS to determine the sentence of the defendant, Loomis argued that the secretive inner workings of the algorithm

violated his due process rights. Though the Wisconsin Supreme Court found that the sentence given would be no different without COMPAS in this case, it did issue warnings about the algorithm's accuracy and risks to minority defendants. Other researchers have questioned whether it is appropriate to use algorithms in applications such as sentencing.

We could hope for an algorithm that is both well calibrated and equal opportunity, but, as Kleinberg *et al.* (2016) show, that is impossible. If the base classes are different, then any algorithm that is well calibrated will necessarily not provide equal opportunity, and vice versa. How can we weigh the two criteria? Equal impact is one possibility. In the case of COMPAS, this means weighing the negative utility of defendants being falsely classified as high risk and losing their freedom, versus the cost to society of an additional crime being committed, and finding the point that optimizes the tradeoff. This is complicated because there are multiple costs to consider. There are individual costs—a defendant who is wrongfully held in jail suffers a loss, as does the victim of a defendant who was wrongfully released and re-offends. But beyond that there are group costs—everyone has a certain fear that they will be wrongfully jailed, or will be the victim of a crime, and all taxpayers contribute to the costs of jails and courts. If we give value to those fears and costs in proportion to the size of a group, then utility for the majority may come at the expense of a minority.

Another problem with the whole idea of recidivism scoring, regardless of the model used, is that we don't have unbiased ground truth data. The data does not tell us who has *committed* a crime—all we know is who has been *convicted* of a crime. If the arresting officers, judge, or jury is biased, then the data will be biased. If more officers patrol some locations, then the data will be biased against people in those locations. Only defendants who are released are candidates to recommit, so if the judges making the release decisions are biased, the data may be biased. If you assume that behind the biased data set there is an underlying, unknown, unbiased data set which has been corrupted by an agent with biases, then there are techniques to recover an approximation to the unbiased data. Jiang and Nachum (2019) describe various scenarios and the techniques involved.

One more risk is that machine learning can be used to *justify* bias. If decisions are made by a biased human after consulting with a machine learning system, the human can say "here is how my interpretation of the model supports my decision, so you shouldn't question my decision." But other interpretations could lead to an opposite decision.

Sometimes fairness means that we should reconsider the objective function, not the data or the algorithm. For example, in making job hiring decisions, if the objective is to hire candidates with the best qualifications in hand, we risk unfairly rewarding those who have had advantageous educational opportunities throughout their lives, thereby enforcing class boundaries. But if the objective is to hire candidates with the best ability to learn on the job, we have a better chance to cut across class boundaries and choose from a broader pool. Many companies have programs designed for such applicants, and find that after a year of training, the employees hired this way do as well as the traditional candidates. Similarly, just 18% of computer science graduates in the U.S. are women, but some schools, such as Harvey Mudd University, have achieved 50% parity with an approach that is focused on encouraging and retaining those who start the computer science program, especially those who start with less programming experience.

A final complication is deciding which classes deserve protection. In the U.S., the Fair Housing Act recognized seven protected classes: race, color, religion, national origin, sex,

disability, and familial status. Other local, state, and federal laws recognize other classes, including sexual orientation, and pregnancy, marital, and veteran status. Is it fair that these classes count for some laws and not others? International human rights law, which encompasses a broad set of protected classes, is a potential framework to harmonize protections across various groups.

Even in the absence of societal bias, **sample size disparity** can lead to biased results. In most data sets there will be fewer training examples of minority class individuals than of majority class individuals. Machine learning algorithms give better accuracy with more training data, so that means that members of minority classes will experience lower accuracy. For example, Buolamwini and Gebru (2018) examined a computer vision gender identification service, and found that it had near-perfect accuracy for light-skinned males, and a 33% error rate for dark-skinned females. A constrained model may not be able to simultaneously fit both the majority and minority class—a linear regression model might minimize average error by fitting just the majority class, and in an SVM model, the support vectors might all correspond to majority class members.

Sample size disparity

Bias can also come into play in the software development process (whether or not the software involves machine learning). Engineers who are debugging a system are more likely to notice and fix those problems that are applicable to themselves. For example, it is difficult to notice that a user interface design won't work for colorblind people unless you are in fact colorblind, or that an Urdu language translation is faulty if you don't speak Urdu.

How can we defend against these biases? First, understand the limits of the data you are using. It has been suggested that data sets (Gebru *et al.*, 2018; Hind *et al.*, 2018) and models (Mitchell *et al.*, 2019) should come with annotations: declarations of provenance, security, conformity, and fitness for use. This is similar to the **data sheets** that accompany electronic components such as resistors; they allow designers to decide what components to use. In addition to the data sheets, it is important to train engineers to be aware of issues of fairness and bias, both in school and with on-the-job training. Having a diversity of engineers from different backgrounds makes it easier for them to notice problems in the data or models. A study by the AI Now Institute (West *et al.*, 2019) found that only 18% of authors at leading AI conferences and 20% of AI professors are women. Black AI workers are at less than 4%. Rates at industry research labs are similar. Diversity could be increased by programs earlier in the pipeline—in college or high school—and by greater awareness at the professional level. Joy Buolamwini founded the Algorithmic Justice League to raise awareness of this issue and develop practices for accountability.

Data sheet

A second idea is to de-bias the data (Zemel *et al.*, 2013). We could over-sample from minority classes to defend against sample size disparity. Techniques such as SMOTE, the synthetic minority over-sampling technique (Chawla *et al.*, 2002) or ADASYN, the adaptive synthetic sampling approach for imbalanced learning (He *et al.*, 2008), provide principled ways of oversampling. We could examine the provenance of data and, for example, eliminate examples from judges who have exhibited bias in their past court cases. Some analysts object to the idea of discarding data, and instead would recommend building a hierarchical model of the data that includes sources of bias, so they can be modeled and compensated for. Google and NeurIPS have attempted to raise awareness of this issue by sponsoring the Inclusive Images Competition, in which competitors train a network on a data set of labeled images collected in North America and Europe, and then test it on images taken from all around the

world. The issue is that given this data set, it is easy to apply the label "bride" to a woman in a standard Western wedding dress, but harder to recognize traditional African and Indian matrimonial dress.

A third idea is to invent new machine learning models and algorithms that are more resistant to bias; and the final idea is to let a system make initial recommendations that may be biased, but then train a second system to de-bias the recommendations of the first one. Bellamy *et al.* (2018) introduced the IBM AI FAIRNESS 360 system, which provides a framework for all of these ideas. We expect there will be increased use of tools like this in the future.

How do you make sure that the systems you build will be fair? A set of best practices has been emerging (although they are not always followed):

- Make sure that the software engineers talk with social scientists and domain experts to understand the issues and perspectives, and consider fairness from the start.
- Create an environment that fosters the development of a diverse pool of software engineers that are representative of society.
- Define what groups your system will support: different language speakers, different age groups, different abilities with sight and hearing, etc.
- Optimize for an objective function that incorporates fairness.
- Examine your data for prejudice and for correlations between protected attributes and other attributes.
- Understand how any human annotation of data is done, design goals for annotation accuracy, and verify that the goals are met.
- Don't just track overall metrics for your system; make sure you track metrics for subgroups that might be victims of bias.
- Include system tests that reflect the experience of minority group users.
- Have a feedback loop so that when fairness problems come up, they are dealt with.

27.3.4 Trust and transparency

Trust

It is one challenge to make an AI system accurate, fair, safe, and secure; a different challenge to convince everyone else that you have done so. People need to be able to **trust** the systems they use. A PwC survey in 2017 found that 76% of businesses were slowing the adoption of AI because of trustworthiness concerns. In Section 19.9.4 we covered some of the engineering approaches to trust; here we discuss the policy issues.

Verification and validation

To earn trust, any engineered systems must go through a **verification and validation** (V&V) process. Verification means that the product satisfies the specifications. Validation means ensuring that the specifications actually meet the needs of the user and other affected parties. We have an elaborate V&V methodology for engineering in general, and for traditional software development done by human coders; much of that is applicable to AI systems. But machine learning systems are different and demand a different V&V process, which has not yet been fully developed. We need to verify the data that these systems learn from; we need to verify the accuracy and fairness of the results, even in the face of uncertainty that makes an exact result unknowable; and we need to verify that adversaries cannot unduly influence the model, nor steal information by querying the resulting model.

Certification

One instrument of trust is **certification**; for example, Underwriters Laboratories (UL) was founded in 1894 at a time when consumers were apprehensive about the risks of electric

power. UL certification of appliances gave consumers increased trust, and in fact UL is now considering entering the business of product testing and certification for AI.

Other industries have long had safety standards. For example, ISO 26262 is an international standard for the safety of automobiles, describing how to develop, produce, operate, and service vehicles in a safe way. The AI industry is not yet at this level of clarity, although there are some frameworks in progress, such as IEEE P7001, a standard defining ethical design for artificial intelligence and autonomous systems (Bryson and Winfield, 2017). There is ongoing debate about what kind of certification is necessary, and to what extent it should be done by the government, by professional organizations like IEEE, by independent certifiers such as UL, or through self-regulation by the product companies.

Another aspect of trust is **transparency**: consumers want to know what is going on inside a system, and that the system is not working against them, whether due to intentional malice, an unintentional bug, or pervasive societal bias that is recapitulated by the system. In some cases this transparency is delivered directly to the consumer. In other cases their are intellectual property issues that keep some aspects of the system hidden to consumers, but open to regulators and certification agencies.

Transparency

When an AI system turns you down for a loan, you deserve an explanation. In Europe, the GDPR enforces this for you. An AI system that can explain itself is called **explainable AI (XAI)**. A good explanation has several properties: it should be understandable and convincing to the user, it should accurately reflect the reasoning of the system, it should be complete, and it should be specific in that different users with different conditions or different outcomes should get different explanations.

Explainable AI (XAI)

It is quite easy to give a decision algorithm access to its own deliberative processes, simply by recording them and making them available as data structures. This means that machines may eventually be able to give better explanations of their decisions than humans can. Moreover, we can take steps to certify that the machine's explanations are not deceptions (intentional or self-deception), something that is more difficult with a human.

An explanation is a helpful but not sufficient ingredient to trust. One issue is that explanations are not decisions: they are stories about decisions. As discussed in Section 19.9.4, we say that a system is interpretable if we can inspect the source code of the model and see what it is doing, and we say it is explainable if we can make up a story about what it is doing—even if the system itself is an uninterpretable black box. To explain an uninterpretable black box, we need to build, debug, and test a separate explanation system, and make sure it is in sync with the original system. And because humans love a good story, we are all too willing to be swayed by an explanation that sounds good. Take any political controversy of the day, and you can always find two so-called experts with diametrically opposed explanations, both of which are internally consistent.

A final issue is that an explanation about one case does not give you a summary over other cases. If the bank explains, "Sorry, you didn't get the loan because you have a history of previous financial problems," you don't know if that explanation is accurate or if the bank is secretly biased against you for some reason. In this case, you require not just an explanation, but also an **audit** of past decisions, with aggregated statistics across various demographic groups, to see if their approval rates are balanced.

Part of transparency is knowing whether you are interacting with an AI system or a human. Toby Walsh (2015) proposed that "an autonomous system should be designed so that

it is unlikely to be mistaken for anything besides an autonomous system, and should identify itself at the start of any interaction." He called this the "red flag" law, in honor of the UK's 1865 Locomotive Act, which required any motorized vehicle to have a person with a red flag walk in front of it, to signal the oncoming danger.

In 2019, California enacted a law stating that "It shall be unlawful for any person to use a bot to communicate or interact with another person in California online, with the intent to mislead the other person about its artificial identity."

27.3.5 The future of work

From the first agricultural revolution (10,000 BCE) to the industrial revolution (late 18th century) to the green revolution in food production (1950s), new technologies have changed the way humanity works and lives. A primary concern arising from the advance of AI is that human labor will become obsolete. Aristotle, in Book I of his *Politics*, presents the main point quite clearly:

> For if every instrument could accomplish its own work, obeying or anticipating the will of others . . . if, in like manner, the shuttle would weave and the plectrum touch the lyre without a hand to guide them, chief workmen would not want servants, nor masters slaves.

Everyone agrees with Aristotle's observation that there is an immediate reduction in employment when an employer finds a mechanical method to perform work previously done by a person. The issue is whether the so-called compensation effects that ensue—and that tend to increase employment—will eventually make up for this reduction. The primary compensation effect is the increase in overall wealth from greater productivity, which leads in turn to greater demand for goods and tends to increase employment. For example, PwC (Rao and Verweij, 2017) predicts that AI contribute $15 trillion annually to global GDP by 2030. The healthcare and automotive/transportation industries stand to gain the most in the short term. However, the advantages of automation have not yet taken over in our economy: the current rate of growth in labor productivity is actually below historical standards. Brynjolfsson *et al.* (2018) attempt to explain this paradox by suggesting that the lag between the development of basic technology and its implementation in the economy is longer than commonly supposed.

Technological innovations have historically put some people out of work. Weavers were replaced by automated looms in the 1810s, leading to the Luddite protests. The Luddites were not against technology *per se*; they just wanted the machines to be used by skilled workers paid a good wage to make high-quality goods, rather than by unskilled workers to make poor-quality goods at low wages. The global destruction of jobs in the 1930s led John Maynard Keynes to coin the term **technological unemployment**. In both cases, and several others, employment levels eventally recovered.

Technological unemployment

The mainstream economic view for most of the 20th century was that technological employment was at most a short-term phenomenon. Increased productivity would always lead to increased wealth and increased demand, and thus net job growth. A commonly cited example is that of bank tellers: although ATMs replaced humans in the job of counting out cash for withdrawals, that made it cheaper to operate a bank branch, so the number of branches increased, leading to more bank employees overall. The nature of the work also changed, becoming less routine and requiring more advanced business skills. The net effect of automation seems to be in eliminating *tasks* rather than *jobs*.

The majority of commenters predict that the same will hold true with AI technology, at least in the short run. Gartner, McKinsey, Forbes, the World Economic Forum, and the Pew Research Center each released reports in 2018 predicting a net increase in jobs due to AI-driven automation. But some analysts think that this time around, things will be different. In 2019, IBM predicted that 120 million workers would need retraining due to automation by 2022, and Oxford Economics predicted that 20 million manufacturing jobs could be lost to automation by 2030.

Frey and Osborne (2017) survey 702 different occupations, and estimate that 47% of them are at risk of being automated, meaning that at least some of the tasks in the occupation can be performed by machine. For example, almost 3% of the workforce in the U.S. are vehicle drivers, and in some districts, as much as 15% of the male workforce are drivers. As we saw in Chapter 26, the task of driving is likely to be eliminated by driverless cars/trucks/buses/taxis.

It is important to distinguish between occupations and the tasks within those occupations. McKinsey estimates that only 5% of occupations are fully automatable, but that 60% of occupations can have about 30% of their tasks automated. For example, future truck drivers will spend less time holding the steering wheel and more time making sure that the goods are picked up and delivered properly; serving as customer service representatives and salespeople at either end of the journey; and perhaps managing convoys of, say, three robotic trucks. Replacing three drivers with one convoy manager implies a net loss in employment, but if transportation costs decrease, there will be more demand, which wins some of the jobs back—but perhaps not all of them. As another example, despite many advances in applying machine learning to the problem of medical imaging, radiologists have so far been augmented, not replaced, by these tools. Ultimately, there is a choice of how to make use of automation: do we want to focus on *cutting cost*, and thus see job loss as a positive; or do we want to focus on *improving quality*, making life better for the worker and the customer?

It is difficult to predict exact timelines for automation, but currently, and for the next few years, the emphasis is on automation of structured analytical tasks, such as reading x-ray images, customer relationship management (e.g., bots that automatically sort customer complaints and respond with suggested remedies), and **business process automation** that combines text documents and structured data to make business decisions and improve workflow. Over time, we will see more automation with physical robots, first in controlled warehouse environments, then in more uncertain environments, building to a significant portion of the marketplace by around 2030.

Business process automation

As populations in developed countries grow older, the ratio between workers and retirees changes. In 2015 there were less than 30 retirees per 100 workers; by 2050 there may be over 60 per 100 workers. Care for the elderly will be an increasingly important role, one that can partially be filled by AI. Moreover, if we want to maintain the current standard of living, it will also be necessary to make the remaining workers more productive; automation seems like the best opportunity to do that.

Even if automation has a multi-trillion-dollar net positive impact, there may still be problems due to the **pace of change**. Consider how change came to the farming industry: in 1900, over 40% of the U.S. workforce was in agriculture, but by 2000 that had fallen to 2%.[3] That

Pace of change

[3] In 2010, although only 2% of the U.S. workforce were actual farmers, over 25% of the population (80 million people) played the FARMVILLE game at least once.

is a huge disruption in the way we work, but it happened over a period of 100 years, and thus across generations, not in the lifetime of one worker.

Workers whose jobs are automated away this decade may have to retrain for a new profession within a few years—and then perhaps see their new profession automated and face yet another retraining period. Some may be happy to leave their old profession—we see that as the economy improves, trucking companies need to offer new incentives to hire enough drivers—but workers will be apprehensive about their new roles. To handle this, we as a society need to provide lifelong education, perhaps relying in part on online education driven by artificial intelligence (Martin, 2012). Bessen (2015) argues that workers will not see increases in income until they are trained to implement the new technologies, a process that takes time.

Income inequality

Technology tends to magnify **income inequality**. In an information economy marked by high-bandwidth global communication and zero-marginal-cost replication of intellectual property (what Frank and Cook (1996) call the "Winner-Take-All Society"), rewards tend to be concentrated. If farmer Ali is 10% better than farmer Bo, then Ali gets about 10% more income: Ali can charge slightly more for superior goods, but there is a limit on how much can be produced on the land, and how far it can be shipped. But if software app developer Cary is 10% better than Dana, it may be that Cary ends up with 99% of the global market. AI increases the pace of technological innovation and thus contributes to this overall trend, but AI also holds the promise of allowing us to take some time off and let our automated agents handle things for a while. Tim Ferriss (2007) recommends using automation and outsourcing to achieve a four-hour work week.

Before the industrial revolution, people worked as farmers or in other crafts, but didn't report to a **job** at a place of work and put in hours for an employer. But today, most adults in developed countries do just that, and the job serves three purposes: it fuels the production of the goods that society needs to flourish, it provides the income that the worker needs to live, and it gives the worker a sense of purpose, accomplishment, and social integration. With increasing automation, it may be that these three purposes become disaggregated—society's needs will be served in part by automation, and in the long run, individuals will get their sense of purpose from contributions other than work. Their income needs can be served by social policies that include a combination of free or inexpensive access to social services and education, portable health care, retirement, and education accounts, progressive tax rates, earned income tax credits, negative income tax, or universal basic income.

27.3.6 Robot rights

The question of robot consciousness, discussed in Section 27.2, is critical to the question of what rights, if any, robots should have. If they have no consciousness, no qualia, then few would argue that they deserve rights.

But if robots can feel pain, if they can dread death, if they are considered "persons," then the argument can be made (e.g., by Sparrow (2004)) that they have rights and deserve to have their rights recognized, just as slaves, women, and other historically oppressed groups have fought to have their rights recognized. The issue of robot personhood is often considered in fiction: from Pygmalion to Coppélia to Pinocchio to the movies *AI* and *Centennial Man*, we have the legend of a doll/robot coming to life and striving to be accepted as a human with human rights. In real life, Saudi Arabia made headlines by giving honorary citizenship to Sophia, a human-looking puppet capable of speaking preprogrammed lines.

If robots have rights, then they should not be enslaved, and there is a question of whether reprogramming them would be a kind of enslavement. Another ethical issue involves voting rights: a rich person could buy thousands of robots and program them to cast thousands of votes—should those votes count? If a robot clones itself, can they both vote? What is the boundary between ballot stuffing and exercising free will, and when does robotic voting violate the "one person, one vote" principle?

Ernie Davis argues for avoiding the dilemmas of robot consciousness by never building robots that could possibly be considered conscious. This argument was previously made by Joseph Weizenbaum in his book *Computer Power and Human Reason* (1976), and before that by Julien de La Mettrie in *L'Homme Machine* (1748). Robots are tools that we create, to do the tasks we direct them to do, and if we grant them personhood, we are just declining to take responsibility for the actions of our own property: "I'm not at fault for my self-driving car crash—the car did it itself."

This issue takes a different turn if we develop human–robot hybrids. Of course we already have humans enhanced by technology such as contact lenses, pacemakers, and artificial hips. But adding computational protheses may blur the lines between human and machine.

27.3.7 AI Safety

Almost any technology has the potential to cause harm in the wrong hands, but with AI and robotics, the hands might be operating on their own. Countless science fiction stories have warned about robots or cyborgs running amok. Early examples include Mary Shelley's *Frankenstein, or the Modern Prometheus* (1818) and Karel Čapek's play *R.U.R.* (1920), in which robots conquer the world. In movies, we have *The Terminator* (1984) and *The Matrix* (1999), which both feature robots trying to eliminate humans—the **robopocalypse** (Wilson, 2011). Perhaps robots are so often the villains because they represent the unknown, just like the witches and ghosts of tales from earlier eras. We can hope that a robot that is smart enough to figure out how to terminate the human race is also smart enough to figure out that that was not the intended utility function; but in building intelligent systems, we want to rely not just on hope, but on a design process with guarantees of safety.

Robopocalypse

It would be unethical to distribute an unsafe AI agent. We require our agents to avoid accidents, to be resistant to adversarial attacks and malicious abuse, and in general to cause benefits, not harms. That is especially true as AI agents are deployed in safety-critical applications, such as driving cars, controlling robots in dangerous factory or construction settings, and making life-or-death medical decisions.

There is a long history of **safety engineering** in traditional engineering fields. We know how to build bridges, airplanes, spacecraft, and power plants that are designed up front to behave safely even when components of the system fail. The first technique is **failure modes and effect analysis (FMEA)**: analysts consider each component of the system, and imagine every possible way the component could go wrong (for example, what if this bolt were to snap?), drawing on past experience and on calculations based on the physical properties of the component. Then the analysts work forward to see what would result from the failure. If the result is severe (a section of the bridge could fall down) then the analysts alter the design to mitigate the failure. (With this additional cross-member, the bridge can survive the failure of any 5 bolts; with this backup server, the online service can survive a tsunami taking out the primary server.) The technique of **fault tree analysis (FTA)** is used to make these

Safety engineering

Failure modes and effect analysis (FMEA)

Fault tree analysis (FTA)

determinations: analysts build an AND/OR tree of possible failures and assign probabilities to each root cause, allowing for calculations of overall failure probability. These techniques can and should be applied to all safety-critical engineered systems, including AI systems.

The field of **software engineering** is aimed at producing reliable software, but the emphasis has historically been on *correctness*, not *safety*. Correctness means that the software faithfully implements the specification. But safety goes beyond that to insist that the specification has considered any feasible failure modes, and is designed to degrade gracefully even in the face of unforeseen failures. For example, the software for a self-driving car wouldn't be considered safe unless it can handle unusual situations. For example, what if the power to the main computer dies? A safe system will have a backup computer with a separate power supply. What if a tire is punctured at high speed? A safe system will have tested for this, and will have software to correct for the resulting loss of control.

Unintended side effect

An agent designed as a utility maximizer, or as a goal achiever, can be unsafe if it has the wrong objective function. Suppose we give a robot the task of fetching a coffee from the kitchen. We might run into trouble with **unintended side effects**—the robot might rush to accomplish the goal, knocking over lamps and tables along the way. In testing, we might notice this kind of behavior and modify the utility function to penalize such damage, but it is difficult for the designers and testers to anticipate *all* possible side effects ahead of time.

Low impact

One way to deal with this is to design a robot to have **low impact** (Armstrong and Levinstein, 2017): instead of just maximizing utility, maximize the utility minus a weighted summary of all changes to the state of the world. In this way, all other things being equal, the robot prefers not to change those things whose effect on utility is unknown; so it avoids knocking over the lamp not because it knows specifically that knocking the lamp will cause it to fall over and break, but because it knows in general that disruption might be bad. This can be seen as a version of the physician's creed "first, do no harm," or as an analog to **regularization** in machine learning: we want a policy that achieves goals, but we prefer policies that take smooth, low-impact actions to get there. The trick is how to measure impact. It is not acceptable to knock over a fragile lamp, but perfectly fine if the air molecules in the room are disturbed a little, or if some bacteria in the room are inadvertently killed. It is certainly not acceptable to harm pets and humans in the room. We need to make sure that the robot knows the differences between these cases (and many subtle cases in between) through a combination of explicit programming, machine learning over time, and rigorous testing.

Utility functions can go wrong due to **externalities**, the word used by economists for factors that are outside of what is measured and paid for. The world suffers when greenhouse gases are considered as externalities—companies and countries are not penalized for producing them, and as a result everyone suffers. Ecologist Garrett Hardin (1968) called the exploitation of shared resources the **tragedy of the commons**. We can mitigate the tragedy by internalizing the externalities—making them part of the utility function, for example with a carbon tax—or by using the design principles that economist Elinor Ostrom identified as being used by local people throughout the world for centuries (work that won her the Nobel Prize in Economics in 2009):

- Clearly define the shared resource and who has access.
- Adapt to local conditions.
- Allow all parties to participate in decisions.

- Monitor the resource with accountable monitors.
- Sanctions, proportional to the severity of the violation.
- Easy conflict resolution procedures.
- Hierarchical control for large shared resources.

Victoria Krakovna (2018) has cataloged examples of AI agents that have gamed the system, figuring out how to maximize utility without actually solving the problem that their designers intended them to solve. To the designers this looks like cheating, but to the agents, they are just doing their job. Some agents took advantage of bugs in the simulation (such as floating point overflow bugs) to propose solutions that would not work once the bug was fixed. Several agents in video games discovered ways to crash or pause the game when they were about to lose, thus avoiding a penalty. And in a specification where crashing the game was penalized, one agent learned to use up just enough of the game's memory so that when it was the opponent's turn, it would run out of memory and crash the game. Finally, a genetic algorithm operating in a simulated world was supposed to evolve fast-moving creatures but in fact produced creatures that were enormously tall and moved fast by falling over.

Designers of agents should be aware of these kinds of specification failures and take steps to avoid them. To help them do that, Krakovna was part of the team that released the AI Safety Gridworlds environments (Leike *et al.*, 2017), which allows designers to test how well their agents perform.

The moral is that we need to be very careful in specifying what we want, because with utility maximizers we get what we actually asked for. The **value alignment problem** is the problem of making sure that what we ask for is what we really want; it is also known as the **King Midas problem**, as discussed on page 33. We run into trouble when a utility function fails to capture background societal norms about acceptable behavior. For example, a human who is hired to clean floors, when faced with a messy person who repeatedly tracks in dirt, knows that it is acceptable to politely ask the person to be more careful, but it is not acceptable to kidnap or incapacitate said person.

Value alignment problem

A robotic cleaner needs to know these things too, either through explicit programming or by learning from observation. Trying to write down all the rules so that the robot always does the right thing is almost certainly hopeless. We have been trying to write loophole-free tax laws for several thousand years without success. Better to make the robot *want* to pay taxes, so to speak, than to try to make rules to force it to do so when it really wants to do something else. A sufficiently intelligent robot will find a way to do something else.

Robots can learn to conform better with human preferences by observing human behavior. This is clearly related to the notion of apprenticeship learning (Section 22.6). The robot may learn a policy that directly suggests what actions to take in what situations; this is often a straightforward supervised learning problem if the environment is observable. For example, a robot can watch a human playing chess: each state–action pair is an example for the learning process. Unfortunately, this form of **imitation learning** means that the robot will repeat human mistakes. Instead, the robot can apply **inverse reinforcement learning** to discover the utility function that the humans must be operating under. Watching even terrible chess players is probably enough for the robot to learn the objective of the game. Given just this information, the robot can then go on to exceed human performance—as, for example, ALPHAZERO did in chess—by computing optimal or near-optimal policies from the objec-

tive. This approach works not just in board games, but in real-world physical tasks such as helicopter aerobatics (Coates *et al.*, 2009).

In more complex settings involving, for example, social interactions with humans, it is very unlikely that the robot will converge to exact and correct knowledge of each human's individual preferences. (After all, many humans never quite learn what makes other humans tick, despite a lifetime of experience, and many of us are unsure of our own preferences too.) It will be necessary, therefore, for machines to function appropriately when it is uncertain about human preferences. In Chapter 18, we introduced **assistance games**, which capture exactly this situation. Solutions to assistance games include acting cautiously, so as not to disturb aspects of the world that the human might care about, and asking questions. For example, the robot could ask whether turning the oceans into sulphuric acid is an acceptable solution to global warming before it puts the plan into effect.

In dealing with humans, a robot solving an assistance game must accommodate human imperfections. If the robot asks permission, the human may give it, not foreseeing that the robot's proposal is in fact catastrophic in the long term. Moreover, humans do not have complete introspective access to their true utility function, and they don't always act in a way that is compatible with it. Humans sometimes lie or cheat, or do things they know are wrong. They sometimes take self-destructive actions like overeating or abusing drugs. AI systems need not learn to adopt these problematic tendencies, but they must understand that they exist when interpreting human behavior to get at the underlying human preferences.

Despite this toolbox of safeguards, there is a fear, expressed by prominent technologists such as Bill Gates and Elon Musk and scientists such as Stephen Hawking and Martin Rees, that AI could evolve out of control. They warn that we have no experience controlling powerful nonhuman entities with super-human capabilities. However, that's not quite true; we have centuries of experience with nations and corporations; non-human entities that aggregate the power of thousands or millions of people. Our record of controlling these entities is not very encouraging: nations produce periodic convulsions called wars that kill tens of millions of human beings, and corporations are partly responsible for global warming and our inability to confront it.

AI systems may present much greater problems than nations and corporations because of their potential to self-improve at a rapid pace, as considered by I. J. Good (1965b):

Ultraintelligent machine

> Let an **ultraintelligent machine** be defined as a machine that can far surpass all the intellectual activities of any man however clever. Since the design of machines is one of these intellectual activities, an ultraintelligent machine could design even better machines; there would then unquestionably be an "intelligence explosion," and the intelligence of man would be left far behind. Thus the first ultraintelligent machine is the *last* invention that man need ever make, provided that the machine is docile enough to tell us how to keep it under control.

Technological singularity

Good's "intelligence explosion" has also been called the **technological singularity** by mathematics professor and science fiction author Vernor Vinge, who wrote in 1993: "Within thirty years, we will have the technological means to create superhuman intelligence. Shortly after, the human era will be ended." In 2017, inventor and futurist Ray Kurzweil predicted the singularity would appear by 2045, which means it got 2 years closer in 24 years. (At that rate, only 336 years to go!) Vinge and Kurzweil correctly note that technological progress on many measures is growing exponentially at present.

It is, however, quite a leap to extrapolate all the way from the rapidly decreasing cost of computation to a singularity. So far, every technology has followed an S-shaped curve, where the exponential growth eventually tapers off. Sometimes new technologies step in when the old ones plateau, but sometimes it is not possible to keep the growth going, for technical, political, or sociological reasons. For example, the technology of flight advanced dramatically from the Wright brothers' flight in 1903 to the moon landing in 1969, but has had no breakthroughs of comparable magnitude since then.

Another obstacle in the way of ultraintelligent machines taking over the world is the world. More specifically, some kinds of progress require not just thinking but acting in the physical world. (Kevin Kelly calls the overemphasis on pure intelligence **thinkism**.) An ultraintelligent machine tasked with creating a grand unified theory of physics might be capable of cleverly manipulating equations a billion times faster than Einstein, but to make any real progress, it would still need to raise millions of dollars to build a more powerful supercollider and run physical experiments over the course of months or years. Only then could it start analyzing the data and theorizing. Depending on how the data turn out, the next step might require raising additional billions of dollars for an interstellar probe mission that would take centuries to complete. The "ultraintelligent thinking" part of this whole process might actually be the least important part. As another example, an ultraintelligent machine tasked with bringing peace to the Middle East might just end up getting 1000 times more frustrated than a human envoy. As yet, we don't know how many of the big problems are like mathematics and how many are like the Middle East.

Thinkism

While some people fear the singularity, others relish it. The **transhumanism** social movement looks forward to a future in which humans are merged with—or replaced by—robotic and biotech inventions. Ray Kurzweil writes in *The Singularity is Near* (2005):

Transhumanism

> The Singularity will allow us to transcend these limitations of our biological bodies and brain. We will gain power over our fates. ... We will be able to live as long as we want ... We will fully understand human thinking and will vastly extend and expand its reach. By the end of this century, the nonbiological portion of our intelligence will be trillions of trillions of times more powerful than unaided human intelligence.

Similarly, when asked whether robots will inherit the Earth, Marvin Minsky said "yes, but they will be our children." These possibilities present a challenge for most moral theorists, who take the preservation of human life and the human species to be a good thing. Kurzweil also notes the potential dangers, writing "But the Singularity will also amplify the ability to act on our destructive inclinations, so its full story has not yet been written." We humans would do well to make sure that any intelligent machine we design today that might evolve into an ultraintelligent machine will do so in a way that ends up treating us well. As Eric Brynjolfsson puts it, "The future is not preordained by machines. It's created by humans."

Summary

This chapter has addressed the following issues:

- Philosophers use the term **weak AI** for the hypothesis that machines could possibly behave intelligently, and **strong AI** for the hypothesis that such machines would count as having actual minds (as opposed to simulated minds).

- Alan Turing rejected the question "Can machines think?" and replaced it with a behavioral test. He anticipated many objections to the possibility of thinking machines. Few AI researchers pay attention to the Turing test, preferring to concentrate on their systems' performance on practical tasks, rather than the ability to imitate humans.
- Consciousness remains a mystery.
- AI is a powerful technology, and as such it poses potential dangers, through lethal autonomous weapons, security and privacy breaches, unintended side effects, unintentional errors, and malignant misuse. Those who work with AI technology have an ethical imperative to responsibly reduce those dangers.
- AI systems must be able to demonstrate they are fair, trustworthy, and transparent.
- There are multiple aspects of fairness, and it is impossible to maximize all of them at once. So a first step is to decide what counts as fair.
- Automation is already changing the way people work. As a society, we will have to deal with these changes.

Bibliographical and Historical Notes

Weak AI: When Alan Turing (1950) proposed the possibility of AI, he also posed many of the key philosophical questions, and provided possible replies. But various philosophers had raised similar issues long before AI was invented. Maurice Merleau-Ponty's *Phenomenology of Perception* (1945) stressed the importance of the body and the subjective interpretation of reality afforded by our senses, and Martin Heidegger's *Being and Time* (1927) asked what it means to actually be an agent. In the computer age, Alva Noe (2009) and Andy Clark (2015) propose that our brains form a rather minimal representation of the world, use the world itself on a just-in-time basis to maintain the illusion of a detailed internal model, and use props in the world (such as paper and pencil as well as computers) to increase the capabilities of the mind. Pfeifer *et al.* (2006) and Lakoff and Johnson (1999) present arguments for how the body helps shape cognition. Speaking of bodies, Levy (2008), Danaher and McArthur (2017), and Devlin (2018) address the issue of robot sex.

Strong AI: René Descartes is known for his dualistic view of the human mind, but ironically his historical influence was toward mechanism and physicalism. He explicitly conceived of animals as automata, and he anticipated the Turing test, writing "it is not conceivable [that a machine] should produce different arrangements of words so as to give an appropriately meaningful answer to whatever is said in its presence, as even the dullest of men can do" (Descartes, 1637). Descartes's spirited defense of the animals-as-automata viewpoint actually had the effect of making it easier to conceive of humans as automata as well, even though he himself did not take this step. The book *L'Homme Machine* (La Mettrie, 1748) did explicitly argue that humans are automata. As far back as Homer (circa 700 BCE), the Greek legends envisioned automata such as the bronze giant Talos and considered the issue of *biotechne*, or life through craft (Mayor, 2018).

The **Turing test** (Turing, 1950) has been debated (Shieber, 2004), anthologized (Epstein *et al.*, 2008), and criticized (Shieber, 1994; Ford and Hayes, 1995). Bringsjord (2008) gives advice for a Turing test judge, and Christian (2011) for a human contestant. The annual Loebner Prize competition is the longest-running Turing test-like contest; Steve Worswick's

MITSUKU won four in a row from 2016 to 2019. The **Chinese room** has been debated endlessly (Searle, 1980; Chalmers, 1992; Preston and Bishop, 2002). Hernández-Orallo (2016) gives an overview of approaches to measuring AI progress, and Chollet (2019) proposes a measure of intelligence based on skill-acquisition efficiency.

Consciousness remains a vexing problem for philosophers, neuroscientists, and anyone who has pondered their own existence. Block (2009), Churchland (2013) and Dehaene (2014) provide overviews of the major theories. Crick and Koch (2003) add their expertise in biology and neuroscience to the debate, and Gazzaniga (2018) shows what can be learned from studying brain disabilities in hospital cases. Koch (2019) gives a theory of consciousness—"intelligence is about doing while experience is about being"—that includes most animals, but not computers. Giulio Tononi and his colleagues propose **integrated information theory** (Oizumi *et al.*, 2014). Damasio (1999) has a theory based on three levels: emotion, feeling, and feeling a feeling. Bryson (2012) shows the value of conscious attention for the process of learning action selection.

The philosophical literature on minds, brains, and related topics is large and jargon-filled. The *Encyclopedia of Philosophy* (Edwards, 1967) is an impressively authoritative and very useful navigation aid. *The Cambridge Dictionary of Philosophy* (Audi, 1999) is shorter and more accessible, and the online *Stanford Encyclopedia of Philosophy* offers many excellent articles and up-to-date references. The *MIT Encyclopedia of Cognitive Science* (Wilson and Keil, 1999) covers the philosophy, biology, and psychology of mind. There are multiple introductions to the philosophical "AI question" (Haugeland, 1985; Boden, 1990; Copeland, 1993; McCorduck, 2004; Minsky, 2007). *The Behavioral and Brain Sciences*, abbreviated *BBS*, is a major journal devoted to philosophical and scientific debates about AI and neuroscience.

Science fiction writer Isaac Asimov (1942, 1950) was one of the first to address the issue of robot ethics, with his **laws of robotics**:

0. A robot may not harm humanity, or through inaction, allow humanity to come to harm.
1. A robot may not injure a human being or, through inaction, allow a human being to come to harm.
2. A robot must obey orders given to it by human beings, except where such orders would conflict with the First Law.
3. A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.

At first glance, these laws seem reasonable. But the trick is how to implement them. Should a robot allow a human to cross the street, or eat junk food, if the human might conceivably come to harm? In Asimov's story *Runaround* (1942), humans need to debug a robot that is found wandering in a circle, acting "drunk." They work out that the circle defines the locus of points that balance the second law (the robot was ordered to fetch some selenium at the center of the circle) with the third law (there is a danger there that threatens the robot's existence).[4] This suggests that the laws are not logical absolutes, but rather are weighed against each other, with a higher weight for the earlier laws. As this was 1942, before the emergence of

[4] Science fiction writers are in broad agreement that robots are very bad at resolving contradictions. In *2001*, the HAL 9000 computer becomes homicidal due to a conflict in its orders, and in the *Star Trek* episode "I, Mudd," Captain Kirk tells an enemy robot that "Everything Harry tells you is a lie," and Harry says "I am lying." At this, smoke comes out of the robot's head and it shuts down.

digital computers, Asimov was probably thinking of an architecture based on control theory via analog computing.

Weld and Etzioni (1994) analyze Asimov's laws and suggest some ways to modify the planning techniques of Chapter 11 to generate plans that do no harm. Asimov has considered many of the ethical issues around technology; in his 1958 story *The Feeling of Power* he tackles the issue of automation leading to a lapse of human skill—a technician rediscovers the lost art of multiplication—as well as the dilemma of what to do when the rediscovery is applied to warfare.

Norbert Wiener's book *God & Golem, Inc.* (1964) correctly predicted that computers would achieve expert-level performance at games and other tasks, and that specifying what it is that we want would prove to be difficult. Wiener writes:

> While it is always possible to ask for something other than we really want, this possibility is most serious when the process by which we are to obtain our wish is indirect, and the degree to which we have obtained our wish is not clear until the very end. Usually we realize our wishes, insofar as we do actually realize them, by a feedback process, in which we compare the degree of attainment of intermediate goals with our anticipation of them. In this process, the feedback goes through us, and we can turn back before it is too late. If the feedback is built into a machine that cannot be inspected until the final goal is attained, the possibilities for catastrophe are greatly increased. I should very much hate to ride on the first trial of an automobile regulated by photoelectric feedback devices, unless there were somewhere a handle by which I could take over control if I found myself driving smack into a tree.

We summarized **codes of ethics** in the chapter, but the list of organizations that have issued sets of principles is growing rapidly, and now includes Apple, DeepMind, Facebook, Google, IBM, Microsoft, the Organisation for Economic Co-operation and Development (OECD), the United Nations Educational, Scientific and Cultural Organization (UNESCO), the U.S. Office of Science and Technology Policy the Beijing Academy of Artificial Intelligence (BAAI), the Institute of Electrical and Electronics Engineers (IEEE), the Association of Computing Machinery (ACM), the World Economic Forum, the Group of Twenty (G20), OpenAI, the Machine Intelligence Research Institute (MIRI), AI4People, the Centre for the Study of Existential Risk, the Center for Human-Compatible AI, the Center for Humane Technology, the Partnership on AI, the AI Now Institute, the Future of Life Institute, the Future of Humanity Institute, the European Union, and at least 42 national governments. We have the handbook on the *Ethics of Computing* (Berleur and Brunnstein, 2001) and introductions to the topic of AI ethics in book (Boddington, 2017) and survey (Etzioni and Etzioni, 2017a) form. The *Journal of Artificial Intelligence and Law* and *AI and Society* cover ethical issues. We'll now look at some of the individual issues.

Lethal autonomous weapons: P. W. Singer's *Wired for War* (2009) raised ethical, legal, and technical issues around robots on the battlefield. Paul Scharre's *Army of None* (2018), written by one of the authors of current US policy on autonomous weapons, offers a balanced and authoritative view. Etzioni and Etzioni (2017b) address the question of whether artificial intelligence should be regulated; they recommend a pause in the development of lethal autonomous weapons, and an international discussion on the subject of regulation.

Privacy: Latanya Sweeney (Sweeney, 2002b) presents the k-anonymity model and the idea of generalizing fields (Sweeney, 2002a). Achieving k-anonymity with minimal loss of data is an NP-hard problem, but Bayardo and Agrawal (2005) give an approximation algorithm. Cynthia Dwork (2008) describes differential privacy, and in subsequent work gives practical examples of clever ways to apply differential privacy to get better results than the naive approach (Dwork *et al.*, 2014). Guo *et al.* (2019) describe a process for certified data removal: if you train a model on some data, and then there is a request to delete some of the data, this extension of differential privacy lets you modify the model and prove that it does not make use of the deleted data. Ji *et al.* (2014) gives a review of the field of privacy. Etzioni (2004) argues for a balancing of privacy and security; individual rights and community. Fung *et al.* (2018), Bagdasaryan *et al.* (2018) discuss the various attacks on federated learning protocols. Narayanan *et al.* (2011) describe how they were able to de-anonymize the obfuscated connection graph from the 2011 Social Network Challenge by crawling the site where the data was obtained (Flickr), and matching nodes with unusually high in-degree or out-degree between the provided data and the crawled data. This allowed them to gain additional information to win the challenge, and it also allowed them to uncover the true identity of nodes in the data. Tools for user privacy are becoming available; for example, TensorFlow provides modules for federated learning and privacy (McMahan and Andrew, 2018).

Fairness: Cathy O'Neil's book *Weapons of Math Destruction* (2017) describes how various black box machine learning models influence our lives, often in unfair ways. She calls on model builders to take responsibility for fairness, and for policy makers to impose appropriate regulation. Dwork *et al.* (2012) showed the flaws with the simplistic "fairness through unawareness" approach. Bellamy *et al.* (2018) present a toolkit for mitigating bias in machine learning systems. Tramèr *et al.* (2016) show how an adversary can "steal" a machine learning model by making queries against an API, Hardt *et al.* (2017) describe equal opportunity as a metric for fairness. Chouldechova and Roth (2018) give an overview of the frontiers of fairness, and Verma and Rubin (2018) give an exhaustive survey of fairness definitions.

Kleinberg *et al.* (2016) show that, in general, an algorithm cannot be both well-calibrated and equal opportunity. Berk *et al.* (2017) give some additional definitions of types of fairness, and again conclude that it is impossible to satisfy all aspects at once. Beutel *et al.* (2019) give advice for how to put fairness metrics into practice.

Dressel and Farid (2018) report on the COMPAS recidivism scoring model. Christin *et al.* (2015) and Eckhouse *et al.* (2019) discuss the use of predictive algorithms in the legal system. Corbett-Davies *et al.* (2017) show that that there is a tension between ensuring fairness and optimizing public safety, and Corbett-Davies and Goel (2018) discuss the differences between fairness frameworks. Chouldechova (2017) advocates for fair impact: all classes should have the same expected utility. Liu *et al.* (2018a) advocate for a long-term measure of impact, pointing out that, for example, if we change the decision point for approving a loan in order to be more fair in the short run, this could have negative effect in the long run on people who end up defaulting on a loan and thus have their credit score reduced.

Since 2014 there has been an annual conference on Fairness, Accountability, and Transparency in Machine Learning. Mehrabi *et al.* (2019) give a comprehensive survey of bias and fairness in machine learning, cataloging 23 kinds of bias and 10 definitions of fairness.

Trust: Explainable AI was an important topic going back to the early days of expert systems (Neches *et al.*, 1985), and has been making a resurgence in recent years (Biran and

Cotton, 2017; Miller *et al.*, 2017; Kim, 2018). Barreno *et al.* (2010) give a taxonomy of the types of security attacks that can be made against a machine learning system, and Tygar (2011) surveys adversarial machine learning. Researchers at IBM have a proposal for gaining trust in AI systems through declarations of conformity (Hind *et al.*, 2018). DARPA requires explainable decisions for its battlefield systems, and has issued a call for research in the area (Gunning, 2016).

AI safety: The book *Artificial Intelligence Safety and Security* (Yampolskiy, 2018) collects essays on AI safety, both recent and classic, going back to Bill Joy's *Why the Future Doesn't Need Us* (Joy, 2000). The "King Midas problem" was anticipated by Marvin Minsky, who once suggested that an AI program designed to solve the Riemann Hypothesis might end up taking over all the resources of Earth to build more powerful supercomputers. Similarly, Omohundro (2008) foresees a chess program that hijacks resources, and Bostrom (2014) describes the runaway paper clip factory that takes over the world. Yudkowsky (2008) goes into more detail about how to design a **Friendly AI**. Amodei *et al.* (2016) present five practical safety problems for AI systems.

Omohundro (2008) describes the *Basic AI Drives* and concludes, "Social structures which cause individuals to bear the cost of their negative externalities would go a long way toward ensuring a stable and positive future." Elinor Ostrom's *Governing the Commons* (1990) describes practices for dealing with externalities by traditional cultures. Ostrom has also applied this approach to the idea of knowledge as a commons (Hess and Ostrom, 2007).

Ray Kurzweil (2005) proclaimed *The Singularity is Near*, and a decade later Murray Shanahan (2015) gave an update on the topic. Microsoft cofounder Paul Allen countered with *The Singularity isn't Near* (2011). He didn't dispute the possibility of ultraintelligent machines; he just thought it would take more than a century to get there. Rod Brooks is a frequent critic of singularitarianism; he points out that technologies often take longer than predicted to mature, that we are prone to magical thinking, and that exponentials don't last forever (Brooks, 2017).

On the other hand, for every optimistic singularitarian there is a pessimist who fears new technology. The Web site `pessimists.co` shows that this has been true throughout history: for example, in the 1890s people were concerned that the elevator would inevitably cause nausea, that the telegraph would lead to loss of privacy and moral corruption, that the subway would release dangerous underground air and disturb the dead, and that the bicycle—especially the idea of a woman riding one—was the work of the devil.

Hans Moravec (2000) introduces some of the ideas of transhumanism, and Bostrom (2005) gives an updated history. Good's ultraintelligent machine idea was foreseen a hundred years earlier in Samuel Butler's *Darwin Among the Machines* (1863). Written four years after the publication of Charles Darwin's *On the Origins of Species* and at a time when the most sophisticated machines were steam engines, Butler's article envisioned "the ultimate development of mechanical consciousness" by natural selection. The theme was reiterated by George Dyson (1998) in a book of the same title, and was referenced by Alan Turing, who wrote in 1951 "At some stage therefore we should have to expect the machines to take control in the way that is mentioned in Samuel Butler's *Erewhon*" (Turing, 1996).

Robot rights: A book edited by Yorick Wilks (2010) gives different perspectives on how we should deal with artificial companions, ranging from Joanna Bryson's view that robots should serve us as tools, not as citizens, to Sherry Turkle's observation that we already per-

sonify our computers and other tools, and are quite willing to blur the boundaries between machines and life. Wilks also contributed a recent update on his views (Wilks, 2019). The philosopher David Gunkel's book *Robot Rights* (2018) considers four possibilities: *can* robots have rights or not, and *should* they or not? The American Society for the Prevention of Cruelty to Robots (ASPCR) proclaims that "The ASPCR is, and will continue to be, exactly as serious as robots are sentient."

The future of work: In 1888, Edward Bellamy published the best-seller *Looking Backward*, which predicted that by the year 2000, technological advances would led to a utopia where equality is achieved and people work short hours and retire early. Soon after, E. M. Forster took the dystopian view in *The Machine Stops* (1909), in which a benevolent machine takes over the running of a society; things fall apart when the machine inevitably fails. Norbert Wiener's prescient book *The Human Use of Human Beings* (1950) argues for the benefits of automation in freeing people from drudgery while offering more creative work, but also discusses several dangers that we recognize as problems today, particularly the problem of value alignment.

The book *Disrupting Unemployment* (Nordfors *et al.*, 2018) discuss some of the ways that work is changing, opening opportunities for new careers. Erik Brynjolfsson and Andrew McAfee address these themes and more in their books *Race Against the Machine* (2011) and *The Second Machine Age* (2014). Ford (2015) describes the challenges of increasing automation, and West (2018) provides recommendations to mitigate the problems, while MIT's Thomas Malone (2004) shows that many of the same issues were apparent a decade earlier, but at that time were attributed to worldwide communication networks, not to automation.